

Where Should the Adapter Go?

Outlier Dimensions, Rank Allocation and Placement in Low-Rank Adaptation of Small Language Models

Author Name

Department

Institution

City, Country

gowhar390275@gmail.com

Abstract—Activation-guided methods such as EVA decide where a low-rank adapter’s capacity goes by profiling the target task’s activations beforehand. We test whether this helps small language models on biomedical classification when budgets are matched and each configuration’s learning rate is tuned on its own dev set. Across eleven methods on three benchmarks, none beats a tuned uniform LoRA by more than 0.34 F1 (1.0 under HoC’s micro-F1), and no difference survives a multiple-comparison correction. A budget sweep, a 4M–110M ladder, a decoder SLM and a clinical task agree once multiple comparisons are corrected (1,328 runs). The methods differ in stability instead, and for activation-PCA initialisation we can trace the cause. On RoBERTa-base, its leading directions lie on the backbone’s outlier dimensions 77 and 588 at 23–24 of the 24 distinct hidden-state inputs and carry a median 59–73× an average direction’s energy, and budget-matched EVA loses 2 of 5 HoC seeds in fp16 and all 3 in fp32. The decoder’s signature is even stronger. Whitening these directions, deflating a general-domain reference subspace (DRIFT, which nests EVA) or the exact generalised-eigenvector contrast all largely restore stability (whitening most reliably) but add no accuracy we can detect at this power. A random-token reference serves as well as real text, so the reference supplies the backbone’s outliers, not domain knowledge. Neither rank allocation nor placement helps at a matched budget, and halving the budget costs nothing. Where a larger budget helps, allocation still does not. We close with a protocol for evaluating allocation.

Index Terms—parameter-efficient fine-tuning, low-rank adaptation, small language models, outlier dimensions, domain adaptation, biomedical text classification

I. INTRODUCTION

Text classification in specialised domains, such as clinical triage, pharmacovigilance and literature curation, is increasingly deployed under constraints that rule out large models. Patient data cannot leave the institution, inference must run on commodity hardware, and dozens of task-specific classifiers must be served concurrently. Small language models (SLMs), from a few million to a few hundred million parameters, suit this setting well, and parameter-efficient fine-tuning (PEFT) is the natural way to specialise them. A frozen backbone is shared across tasks, while a small adapter, typically well under 1% of the parameters, carries the task-specific update [1], [2].

This raises a question that is simple to state but matters more than it might seem: *given a fixed adapter budget, where*

in the network should the capacity go? Uniform allocation, as in standard LoRA [1], is budget-oblivious, so a productive line of work allocates rank non-uniformly. One family does this *during* training, pruning or growing ranks by importance scores [3]–[5]. A second, faster family does it *before* training, by profiling activation geometry on a small calibration set. EVA initialises adapters from the principal directions of the target activations and redistributes rank by explained variance [6]. CorDA [7] and AIRA [8] weight the weight-matrix SVD by an activation covariance, and TLoRA [9] and Liu et al. [10] score layers by task sensitivity or effective rank.

Whether such allocation helps depends on how it is compared. Methods differ in the learning rates they favour, so a baseline that borrows the new method’s learning rate is not a neutral reference. Moving rank between modules of different width also changes the number of parameters spent, which means that a matched rank is not a matched budget. Lee et al. [11] recently showed that LoRA and nine of its variants reach similar peak performance on decoder LLMs once the learning rate of every method is tuned. Training-free allocation has not been evaluated under the same protocol, and neither has the regime in which a good allocation should matter most, namely small backbones adapted to a specialised domain.

In domain adaptation the question takes a sharper form. Biomedical text is still English. Its dominant activation directions are largely those of general text, which the pretrained model already handles, so profiling a single distribution may spend capacity where none is needed. If allocation is to help, it should target directions that carry substantial target-domain energy *and* are poorly represented in pretraining, and only a contrast between two distributions can reveal such directions. To give activation-guided allocation its best chance, we build that contrast into an instrument rather than a competing method. DRIFT (Domain-Residual Informed Fine-Tuning) deflates the target activation covariance by the principal subspace of a general-domain reference corpus, allocates a global parameter budget over the resulting *drift* spectrum, and initialises each adapter inside the residual subspace. It needs two forward passes and no labels or gradients. Its deflation level τ contains EVA as the special case $\tau = 0$, which isolates

the effect of the contrast from everything else, and its exact counterpart, the generalised eigenvectors of the target and reference covariances, tests the contrast in its principled form.

Allocation should matter most for small models. Aghajanyan et al. [12] show that the intrinsic dimension of fine-tuning *grows* as models shrink. A smaller backbone therefore needs a higher-rank update to reach the same quality, and so has less slack to absorb a misallocated budget. This predicts that careful allocation should pay off more as the backbone gets smaller, and we test the prediction on a controlled parameter ladder. The study is organised around four questions:

- RQ1** Under matched parameter budgets, with every configuration tuned on its own, does any allocation or initialisation scheme beat uniform LoRA?
- RQ2** Why is activation-PCA initialisation fragile, and does removing the cause help?
- RQ3** Which design choice matters (rank allocation, initialisation or placement), and does placement still matter when the budgets are matched?
- RQ4** Does the answer change with the backbone’s size or architecture?

Findings. This is an evaluation study, and its contributions lie in its findings:

- **RQ1:** under matched budgets and per-configuration tuning, no method beats LoRA by more than 0.34 F1, no difference with LoRA survives a Holm correction, and the methods differ in stability rather than accuracy (Section V-A).
- **RQ2:** on RoBERTa-base, the leading directions of activation PCA lie on the backbone’s known outlier dimensions and carry tens to thousands of times the energy of an average direction. Recurring dimensions play the same role in BERT-Base, BERT-Medium and a decoder SLM. Whitening, reference deflation and the generalised eigenvectors form one family of remedies. All of them restore most of the stability, none turns it into accuracy, and the reference corpus matters only through the backbone’s outlier dimensions, since random tokens serve as well as WikiText (Sections V-B and V-E).
- **RQ3:** rank allocation alone slightly hurts, and a stable data-derived initialisation helps at most marginally (Section V-C). Adapter placement produces no reliable change at a matched budget either. The feed-forward advantage reported by He et al. [13] does not appear under our controls, and all-module LoRA at half the budget matches the full budget on every task (Section V-D).
- **RQ4:** with every method tuned on each backbone, no gain over LoRA on the BERT ladder survives correction. EVA’s large deficit on small backbones at a rate tuned for RoBERTa-base shrinks to at most 2.5 points at its own rate. The decoder SmoLM2-360M has the strongest outlier signature and repeats the pattern seen on RoBERTa (Section V-E), and so do clinical notes (Section V-F).

Along the way, we provide **DRIFT**, a training-free two-distribution profiler that nests EVA and isolates the contrast

(Section III), and a protocol for evaluating adapter allocation (Section VI).

II. RELATED WORK

Parameter-efficient fine-tuning. Adapter bottlenecks [14], prompt and prefix tuning [15], [16], sparse bias tuning [17] and rescaling vectors [18] all restrict the update to a small subspace. LoRA [1] is the dominant variant because its update merges into the base weights and leaves the inference cost unchanged. DoRA [19] decomposes the update into magnitude and direction, VeRA [20] shares random projections across layers, and QLoRA [21] combines LoRA with quantisation.

Careful re-evaluation. Lee et al. [11] re-evaluate nine LoRA variants with a separate learning-rate search for each method. They find that all reach similar peak performance on decoder LLMs and trace the differences in their optimal learning rates to the largest eigenvalue of the loss Hessian. Biderman et al. [22] document the accuracy gap that remains between LoRA and full fine-tuning, particularly in domains far from pretraining, which is the regime we target. We extend the first kind of study to training-free rank allocation, matched parameter budgets and small encoders, and we add a mechanistic account of the failure we observe.

Rank allocation. AdaLoRA [3] parameterises the update in SVD form and prunes singular triplets by a sensitivity score during training. SoRA [4] gates ranks with a sparsifying penalty, and DyLoRA [5] trains nested ranks. These methods require a training run and start from an inflated budget. Training-free allocation is more recent. EVA [6] redistributes rank by explained variance ratio, AIRA [8] by layer-wise outlier ratios, TLoRA [9] by a sensitivity metric over $W\Sigma_x$, and Liu et al. [10] by effective rank plus a Fréchet distance under virtual perturbation. All of them use statistics of the target task alone. CorDA [7], in its knowledge-preserving mode, uses a covariance from a second distribution to decide which components to freeze, but to our knowledge **DRIFT** is the first to *allocate rank* by a contrast between two distributions.

Spectral initialisation. PiSSA [23] initialises the adapter from the leading singular directions of W and trains those, MiLoRA [24] uses the trailing ones, and LoRA-GA [25] aligns the initialisation with the first task gradient. CorDA [7] orients the decomposition of W using an input-activation covariance. Unlike these methods, **DRIFT** decomposes the *difference in input geometry* between two corpora, not the weight matrix.

Where and at what scale to adapt. The question of which matrices to adapt goes back to LoRA itself, whose authors compare subsets of the attention projections at a fixed parameter count [1, Sec. 7.1]. He et al. [13] find that modifying feed-forward representations uses added parameters more effectively than modifying attention, although attention-only prefix tuning is strong at very small budgets. Our placement results test that finding under matched budgets and separately tuned learning rates on small biomedical encoders. Two results bear on the adapter scale we keep fixed: rsLoRA argues that $s = \alpha/r$ shrinks updates too fast as the rank grows and proposes $\alpha/\sqrt{r}$ [26], and LoRA+ gives A and B different

learning rates [27]. We keep α/r , as EVA does, and tune the learning rate at every rank instead.

Domain adaptation. Continued pretraining on in-domain text (DAPT/TAPT) [28] and from-scratch domain pretraining [29]–[31] both improve biomedical classification, but they update all parameters and produce a separate backbone for each domain. Our aim is complementary. We want to spend a fixed, small adapter budget as well as possible on a frozen general-domain backbone. Contrasting the second-order statistics of two domains is itself classical. CORAL aligns covariances [32], subspace alignment aligns principal subspaces [33], and common spatial patterns in EEG take the generalised eigenvectors of two covariances to find directions of maximal variance ratio [34], [35]. We use the same contrast to choose an adapter’s input subspace rather than to align features.

Outlier dimensions. A handful of hidden dimensions in pretrained transformers carry very large activations, and disabling them badly damages BERT and RoBERTa [36]. For RoBERTa-base they are dimensions 77 and 588, and their magnitude tracks token frequency [37]. Such dimensions have also been traced to positional information [38]. These rogue dimensions dominate similarity measures, and standardising the representation removes their effect [39]. Structured outliers in feed-forward activations complicate quantisation [40], outlier features emerge systematically with scale [41], and LLMs carry massive activations in fixed dimensions [42]. We show that these dimensions capture the leading directions of activation-PCA adapter initialisation. The whitening we test is the adapter-space analogue of the standardisation remedy.

III. ACTIVATION-GUIDED ALLOCATION

We first recall the principle behind EVA (activation PCA) and then construct the reference contrast that defines DRIFT. The two methods share the budgeted allocation and initialisation described in Sections III-E and III-F.

A. Setup

Let the backbone contain adaptable linear modules $m = 1, \dots, M$ with weights $W_m \in \mathbb{R}^{d_m^{\text{out}} \times d_m^{\text{in}}}$. LoRA replaces W_m by $W_m + s B_m A_m$ with $A_m \in \mathbb{R}^{r_m \times d_m^{\text{in}}}$, $B_m \in \mathbb{R}^{d_m^{\text{out}} \times r_m}$, so module m consumes $r_m c_m$ trainable parameters with $c_m = d_m^{\text{in}} + d_m^{\text{out}}$. Given a global budget B we must choose the ranks $\{r_m\}$ subject to $\sum_m r_m c_m \leq B$, and choose the initialisation of each A_m . For uniform rank r the scale is $s = \alpha/r$. When ranks are redistributed, we keep $s = \alpha/r$ for every module, as EVA’s reference implementation does, so moving rank between modules does not change any module’s effective step size.

B. Activation PCA: the adapter subspace is set by input geometry

For a linear module $y = Wx$ and any loss L , the gradient $\nabla_W L = \sum_i \delta_i x_i^\top$, with $\delta_i = \partial L / \partial y_i$, has its row space in $\text{span}\{x_i\}$, and so does every gradient-descent update that starts from $\Delta W = 0$. The input-side factor A should therefore lie in the span of the input activations, and a covariance argument tells us which r directions to pick.

Proposition 1 (Optimal rank- r input subspace): Let $\Delta W^* = \sum_{i=1}^N \delta_i x_i^\top$ with δ_i zero-mean, isotropic ($\mathbb{E}[\delta \delta^\top] = \sigma^2 I$) and independent of x_i , and let $P = A^\top A$ be the orthogonal projector onto the row space of a rank- r factor A . Then

$$\mathbb{E} \|\Delta W^* (I - P)\|_F^2 = \sigma^2 d^{\text{out}} N \text{tr}[(I - P) \Sigma_x], \quad (1)$$

with $\Sigma_x = \frac{1}{N} \sum_i x_i x_i^\top$, and the minimiser over rank- r projectors is the projector onto the top- r eigenvectors of Σ_x [43].

Proposition 1 justifies activation-PCA initialisation [6]. We apply it to the *centred* inputs $x - \mu$, so that Σ_x is the input covariance. The mean μ is common to every input, so an adapter along it contributes only the constant offset $\Delta W \mu$, and in RoBERTa the mean is dominated by a few outlier features [36], [37]. EVA’s reference implementation centres for the same reason. Note, however, that the proposition optimises the *total* gradient energy, which under the isotropic-noise model is dominated by directions where $\|x\|$ is simply large. These need not be domain-specific. They may be the generic directions of the language or, as Section V-B shows, idiosyncratic high-energy dimensions of the backbone.

C. A reference contrast for domain adaptation

Adaptation is driven by the *systematic* part of the gradient, $M = \mathbb{E}_D[\delta x^\top]$, and not by per-example noise, which averages out. This is where the pretrained weights carry information that a single-distribution profile discards. For module m , let P_m^G be the orthogonal projector onto the top- k_m eigenspace of the reference covariance $\Sigma_m^G = \text{Cov}_G[x]$ of a general-domain corpus G (below we drop the index m where a single module is meant). We assume

$$(A) \quad \mathbb{E}_D[\delta (P^G x)^\top] \approx 0, \quad (2)$$

that is, the pretrained module is near-stationary on inputs confined to the reference subspace, as it is on G itself when W is a stationary point of pretraining on G . (A) is a modelling assumption, not a derived result: it states directly that the systematic gradient has no component in the reference subspace. It follows immediately that $M \approx \mathbb{E}_D[\delta ((I - P^G)x)^\top]$, so $\text{row}(M) \subseteq \text{range}(I - P^G)$, and applying Proposition 1 within that range makes the optimal rank- r input basis the leading eigenbasis of the *drift covariance*

$$\tilde{\Sigma} = (I - P^G) \Sigma^D (I - P^G). \quad (3)$$

A pretrained model is not exactly stationary, and no public corpus is its pretraining distribution, so (A) holds only approximately. We therefore treat it as motivation and test its consequence. Section V-B varies only the deflation level τ ($\tau = 0$ removes the contrast) and replaces the reference corpus with controls.

D. One contrast, three approximations

Three ways of taming the target’s dominant directions appear in this paper, and all three approximate a single operation. That operation whitens the target covariance by a reference covariance R and takes its leading directions, i.e. it solves the generalised eigenproblem $\Sigma^D v = \mu R v$.

With $R = \Sigma^G$, the leading v maximise the ratio of target to reference energy, $v^\top \Sigma^D v / v^\top \Sigma^G v$. This is the criterion of common spatial patterns [34], [35] and the trace-ratio form of Fisher’s discriminant, and it automatically suppresses directions that are energetic in both distributions, such as a backbone’s outlier dimensions. DRIFT’s deflation replaces $(\Sigma^G)^{-1}$ by a hard step, with zero weight on the reference’s top- k eigenspace and unit weight elsewhere. Whitened EVA (Section IV) keeps the target’s own principal directions and rescales each to the energy of an average direction. In other words, it whitens them by the target covariance itself. We also run the exact criterion, **GEV**, in which each A_m holds the leading generalised eigenvectors of (Σ_m^D, Σ_m^G) , normalised to unit length. To keep the pencil well conditioned, Σ_m^G is shrunk towards a scaled identity (weight 0.1) [44], and to isolate the choice of subspace, GEV uses uniform rank. If even the exact ratio criterion does not beat LoRA, the negative result does not hinge on how the contrast is approximated.

E. Budgeted allocation

Let $\tilde{\lambda}_{m,1} \geq \tilde{\lambda}_{m,2} \geq \dots$ be the eigenvalues of $\tilde{\Sigma}_m$ and define the normalised drift spectrum $\hat{\lambda}_{m,i} = \tilde{\lambda}_{m,i} / \text{tr}(\Sigma_m^D)$, i.e. the fraction of module m ’s target-domain activation energy that lies outside the reference subspace and is captured by direction i . Normalising by $\text{tr}(\Sigma_m^D)$ makes the score scale-free and therefore comparable across modules whose activations have very different magnitudes. We allocate by maximising total captured drift,

$$\max_{\{r_m\}} \sum_{m=1}^M \sum_{i=1}^{r_m} \hat{\lambda}_{m,i} \quad \text{s.t.} \quad \sum_{m=1}^M r_m c_m \leq B, \quad 0 \leq r_m \leq r_{\max}. \quad (4)$$

Proposition 2 (Greedy allocation): Each per-module objective $V_m(r) = \sum_{i \leq r} \hat{\lambda}_{m,i}$ is concave in r because $\hat{\lambda}_{m,i}$ is non-increasing in i , so (4) is a separable concave resource-allocation problem [45], [46]. Greedy marginal analysis, which repeatedly grants one unit of rank to the module with the largest ratio $\hat{\lambda}_{m,r_m+1} / c_m$ until the best remaining unit no longer fits, is exactly optimal when all c_m are equal. Otherwise, its objective is within the value of a single rank unit, $\max_m \hat{\lambda}_{m,r_m+1}$, of the integer optimum.

Proof sketch. Taking units in decreasing order of value per parameter solves the continuous relaxation, whose optimum consists of the greedy units plus a fraction of the first unit that does not fit. The integer optimum lies between the greedy value and the relaxation’s, which differ by less than the value of that unit. With equal costs, the fraction is empty. $\square$

Greedy marginal analysis therefore allocates the budget near-optimally *for captured drift*, the allocation objective of Eq. (4). Whether that objective tracks accuracy is a separate empirical question, and Section V-C finds that it does not.

F. Initialisation and cost

In practice, P_m^G spans the fewest leading eigenvectors of Σ_m^G that capture a fraction τ of its trace. Each module with $r_m > 0$ receives $A_m = [\tilde{u}_{m,1}, \dots, \tilde{u}_{m,r_m}]^\top$, the leading eigenvectors of $\tilde{\Sigma}_m$, and $B_m = 0$, so that $\Delta W_m = 0$ at initialisation. The

base function is thus preserved exactly, while the input-side factor already points into the drift subspace. Profiling costs two forward passes over about a thousand sequences each, plus symmetric eigendecompositions per module. It needs no labels, no backward pass and no training, and Section V-G measures it at under half of a single LoRA training run.

G. The τ family

The deflation level τ interpolates within a family of methods. At $\tau = 0$ we have $k_m = 0$, $P_m^G = 0$ and $\tilde{\Sigma}_m = \Sigma_m^D$, which gives allocation by the explained variance of the target activations with initialisation from their principal directions, i.e. EVA [6]. As $\tau \rightarrow 1$, the reference subspace fills the space and the drift signal vanishes. Useful values lie strictly in between, and we sweep τ in Section V-B. This nesting attributes any difference from EVA to the deflation alone, with everything else held fixed, and it is what makes DRIFT useful as an instrument even where it is not useful as a method.

IV. EXPERIMENTAL SETUP

Tasks. We use three biomedical classification benchmarks, chosen to differ in granularity, label structure and size. **ChemProt** is 13-way chemical–protein relation classification over PubMed sentences [47] (4,169 / 2,427 / 3,469 train / dev / test), in the split released by Gururangan et al. [28]. It is lexically the furthest from general English. **RCT-20k** is 5-way rhetorical-role classification of sentences in randomised-controlled-trial abstracts [48]. Its signal lies in discourse structure rather than terminology, which makes it a useful negative control for a terminology-driven method. **HoC** is 10-label multi-label classification of abstracts by hallmark of cancer [49] (1,295 / 186 / 371). We reconstruct it at document level from the sentence-level release by grouping on PMID and taking the union of hallmarks, which reproduces the BLURB split exactly [29]. We score HoC with example-based F1. BLURB reports micro-F1, and we report that as well where it allows a comparison with published numbers. We subsample the RCT-20k training set to 5,000 examples and cap its evaluation splits at 6,000, using a fixed seed so that every method sees identical data. ChemProt and HoC are used in full, and ChemProt and RCT-20k are scored with micro-F1. A fourth task, specialty classification of clinical notes, is introduced together with its results in Section V-F.

Backbones. RoBERTa-base (125M) [50] is our primary backbone and lets us compare our numbers with published full fine-tuning and domain-adaptive pretraining results on the same splits [28]. For the scale study we use the BERT miniatures of Turc et al. [51], which range from BERT-Tiny (4.4M) to BERT-Base (110M). They share one vocabulary, one pretraining corpus and one recipe, so only depth and width vary. For the architecture check we use the decoder-only SmolLM2-360M [52], classifying from its last token, with all 224 linear modules adapted.

Adapter placement and budget. All low-rank methods adapt every linear map in every transformer block (the four attention projections plus both feed-forward matrices), giving $M = 72$

modules for RoBERTa-base. The budget is fixed to that of uniform rank 8, i.e. $B = 1,327,104$ adapter parameters (1.06% of the backbone). DRIFT and EVA redistribute rank non-uniformly but never exceed B . Two methods spend slightly more, which, if anything, favours them. DoRA adds a magnitude vector per module (1.41M). AdaLoRA prunes to the same total *rank* as uniform rank 8 but keeps more of it in the costlier feed-forward matrices, so it ends at up to 1.43M after holding $1.5\times$ that during training. We report the parameters actually spent. Every method, including linear probing and BitFit, also trains the backbone’s classification head at the method’s learning rate. For RoBERTa this head is a 768×768 dense layer and an output layer (0.60M parameters), randomly initialised from the run’s seed. It is identical across methods and excluded from the budget and from every parameter count we report. DRIFT uses $\tau = 0.95$, a per-module cap $r_{\max} = \rho r$ with $\rho = 2$ (the same cap EVA uses) and relative scoring.

Methods. We compare full fine-tuning, linear probing, BitFit (all biases) [17], LoRA at uniform rank [1], DoRA [19] and PiSSA spectral initialisation [23]. AdaLoRA uses the cubic budget schedule, sensitivity–uncertainty importance and the orthogonality penalty [3], and is started from $1.5\times$ the target budget and pruned to it. EVA [6] is implemented as in its reference code, with the principal directions of the *centred* target activations, greedy redistribution by explained-variance ratio with cap ρr , and the scale α/r kept fixed after redistribution. We make one change: its greedy step spends our parameter budget B rather than a count of rank units, so that it competes under exactly the same budget. We call this variant *EVA (budget-matched)*, or EVA for short, and we also run EVA’s own rank-unit rule (Section V-B). Because EVA proved unstable, we add its documented whitening option in a scale-normalised form. Each principal row u_i is rescaled by $\sqrt{\bar{\lambda}/\lambda_i}$, with $\bar{\lambda} = \text{tr}(\Sigma^D)/d^{\text{in}}$, so that it responds to the input with the energy of an average direction, and only its *direction* then differs from a random initialisation. Finally, we run DRIFT and GEV (Section III-D). Every low-rank method uses $\alpha = 16$, no adapter dropout, and, unless it prescribes its own initialisation, A drawn Kaiming-uniform and $B = 0$. Every adapter starts from $\Delta W = 0$, so each run starts from the pretrained backbone. All methods are implemented in one codebase (PyTorch 2.10, Transformers 5.0 [53]) and share the data pipeline, optimiser and evaluation, so the only difference is the adapter parameterisation.

Profiles. We compute the target profile from the first 1,024 training texts of each task (taken from the fixed 5,000-example subsample for RCT-20k). The reference profile uses 1,024 WikiText-103 passages [54] of at least 200 characters, drawn with a fixed seed from the 3,227 such passages in its validation and test splits. At $\tau = 0.95$ the reference subspace spans $k_m = 421\text{--}604$ of the 768 dimensions of a hidden-state or attention-output input and $1,412\text{--}2,030$ of the 3,072 feed-forward hidden dimensions, leaving at least 164 residual dimensions per module, ten times the rank cap.

Optimisation. We use AdamW [55] with 6% linear warmup then linear decay, gradient clipping at 1.0 and weight decay

0.01 (none on biases and normalisation weights). Training uses fp16 autocast with dynamic loss scaling, since the T4 has no bf16. The batch size is 32 (16 for HoC and the clinical task) with length-grouped batching, and the maximum sequence length is 128 / 96 / 512 / 512 for ChemProt / RCT / HoC / clinical notes. We train for 10 / 8 / 12 / 10 epochs, respectively, select the epoch by dev-set score, and report the test score at that epoch. The fp32 reruns (Sections V-A and V-E) disable autocast and turn on PyTorch’s deterministic algorithms in warn-only mode, which leaves the memory-efficient attention backward non-deterministic. On the decoder they also use gradient checkpointing, without which fp32 does not fit a T4.

Learning rates. Every configuration from which we draw a conclusion has its own learning rate, selected on the dev set with seed 1. This covers every method of Table I on every task, every adapter placement and budget of Table IV, every budget of the budget sweep, every deflation level τ , every backbone of the ladder, the decoder on both its tasks, GEV, the clinical task, the rsLoRA scale and the linear-head control. Each grid starts from a set of rates suited to the method (e.g. $\{10^{-4}, 3\times 10^{-4}, 10^{-3}\}$ for LoRA, reaching down to 10^{-5} for the profile-initialised methods) and is extended by one half-decade step past whichever edge holds the best dev score until the selection is interior. Appendix A lists every grid and selection. Variants that serve only as controls (the allocation/initialisation factorisation, EVA’s rank-unit rule, the reference controls and the fp32 reruns) inherit the rate of their parent configuration. $\tau = 0.95$ was fixed before any experiment, as the default of our code, and was never tuned, so the τ sweep is descriptive. The test set is never used for any choice.

Reporting. We use three seeds (1, 2, 3), and five (1–5) for LoRA, EVA, whitened EVA and DRIFT, the four methods on which the analysis turns, as well as for the reference controls of Section V-B. A seed fixes the initialisation of the head and of randomly initialised adapters, the data order and the dropout masks, so runs of two methods with the same seed share every nuisance factor except the method. We report the mean and the sample standard deviation (with $n - 1$; at $n = 3$ it is itself noisy) and compare methods with a paired t -test over shared seeds, correcting with Holm’s method [56] over the 24 comparisons with LoRA in Table I. A run is counted as *failed* when its best dev score is more than 10 points below the median of LoRA’s runs on the same task. To make such runs diagnosable rather than merely countable, every run logs, for each epoch, its training loss, the mean and maximum gradient norm before clipping, the number of optimiser steps fp16’s loss scaler skipped and the final loss scale. Section V-A draws on these logs, and they are released with the results. The study comprises 1,328 fine-tuning runs, 443 of them for learning-rate selection, each on one NVIDIA T4.

Availability. All datasets are public. The code, the experiment plans, every per-run result file (with its per-epoch training telemetry) and the profile summaries **will be released with the paper**.

TABLE I

TEST-SET RESULTS WITH A RoBERTA-BASE BACKBONE (MEAN $\pm$ S.D. OVER THREE SEEDS; $\dagger$ FIVE SEEDS). LOW-RANK METHODS ARE HELD TO THE ADAPTER-PARAMETER BUDGET OF UNIFORM RANK 8 (DoRA AND AdaLoRA EXCEED IT SLIGHTLY; THE COLUMN GIVES THE PARAMETERS ACTUALLY SPENT, EXCLUDING THE CLASSIFICATION HEAD THAT EVERY METHOD TRAINS). EVERY METHOD’S LEARNING RATE IS SELECTED ON THE DEV SET FROM A GRID EXTENDED UNTIL THE SELECTION IS INTERIOR (APPENDIX A). HoC MED.: MEDIAN OVER SEEDS. FAILED: RUNS WHOSE BEST DEV SCORE IS MORE THAN 10 POINTS BELOW THE MEDIAN OF LoRA’S RUNS ON THE SAME TASK, OVER ALL THREE TASKS. BEST PARAMETER-EFFICIENT RESULT PER COLUMN IN BOLD, TIES INCLUDED. BELOW THE RULE, THE TWO INSTRUMENTS OF SECTION III-D: DRIFT AND THE EXACT CONTRAST GEV, WHICH KEEPS UNIFORM RANK SO THAT ONLY ITS INITIALISATION DIFFERS FROM LoRA’S.

Method	Adapter params	ChemProt	RCT-20k	HoC	HoC med.	Failed
Full fine-tuning	124.06M	81.7 $\pm$ 0.4	83.7 $\pm$ 0.6	76.2 $\pm$ 1.9	75.5	0/9
Linear probe	0	49.3 $\pm$ 0.6	77.3 $\pm$ 0.6	48.8 $\pm$ 2.0	48.5	–
BitFit	0.10M	81.3 $\pm$ 0.1	83.9 $\pm$ 0.1	78.2$\pm$ 0.6	78.5	0/9
LoRA †	1.33M	82.5 $\pm$ 0.3	83.7 $\pm$ 0.3	78.0 $\pm$ 1.4	78.3	0/15
DoRA	1.41M	82.4 $\pm$ 0.6	83.6 $\pm$ 0.1	57.9 $\pm$ 37.8	79.1	1/9
PiSSA	1.33M	82.4 $\pm$ 0.6	83.8 $\pm$ 0.1	51.2 $\pm$ 33.0	61.5	2/9
AdaLoRA	1.42M	82.7$\pm$ 0.9	82.3 $\pm$ 1.8	78.1 $\pm$ 0.7	77.7	0/9
EVA (budget-matched) †	1.33M	81.5 $\pm$ 1.2	83.7 $\pm$ 0.2	59.4 $\pm$ 25.4	76.5	2/15
EVA (whitened) †	1.33M	82.0 $\pm$ 2.0	84.0$\pm$ 0.1	77.3 $\pm$ 1.3	77.8	0/15
DRIFT †	1.33M	82.4 $\pm$ 0.3	83.9 $\pm$ 0.1	75.7 $\pm$ 8.0	79.0	1/15
GEV	1.33M	82.4 $\pm$ 0.6	83.9 $\pm$ 0.1	77.6 $\pm$ 2.2	78.3	0/9

V. RESULTS

A. RQ1: tuned LoRA is not beaten

Table I summarises the main results. Our training harness reproduces both published reference points available for these splits. Full fine-tuning of RoBERTa-base reaches 81.7 ± 0.4 micro-F1 on ChemProt, against 81.9 ± 1.0 reported by Gururangan et al. [28], and 80.0 ± 2.4 micro-F1 on HoC, against 79.7 reported for RoBERTa-base on the BLURB split [29]. Scoring HoC with micro-F1 instead of example-based F1 changes none of the conclusions below. The only number that moves materially is BitFit’s lead over LoRA on HoC, which widens from 0.3 to 1.0 points and remains non-significant ($p = 0.14$). **No method beats uniform LoRA.** At a matched budget of 1.06% of the backbone, no method improves on LoRA by more than 0.34 F1 on any task (BitFit on HoC, $p = 0.52$). After Holm correction over the 24 comparisons with LoRA, no difference in either direction is significant, and every adjusted p is 1.00. Only one difference reaches $p < 0.05$ uncorrected, and it is a loss: BitFit on ChemProt (-1.1 , $p = 0.04$). We read it, like every nominal result in this paper, as suggestive only. With three to five seeds, the tests rule out large gains but do not establish equivalence. On ChemProt and RCT-20k the parameter-efficient methods lie within 1.7 points of each other. Over five seeds, DRIFT is indistinguishable from LoRA on both tasks (-0.1 , $p = 0.34$; $+0.2$, $p = 0.16$) and from whitened EVA ($p \geq 0.70$). BitFit, with 0.10M parameters, is within 1.4 points of the best method on ChemProt and RCT-20k and is the best method on HoC. The last two rows of the table are the instruments of Section III-D: DRIFT and the exact ratio criterion GEV. At the same budget, GEV is within 0.3 points of LoRA on every task and fails no run. Neither the approximation nor the exact contrast changes this picture.

Stability, not accuracy, separates the methods. HoC, with 1,295 training documents, 512-token inputs and ten labels, is where the methods differ, and the difference lies in failed runs. At its tuned learning rate, PiSSA fails in two of three seeds, DoRA in one, EVA in two of five and DRIFT in one of five (its

TABLE II

HoC IN fp16 (TABLE I) AND RERUN IN fp32 AT THE SAME LEARNING RATES. PyTorch’s DETERMINISTIC ALGORITHMS ON (EXAMPLE-BASED F1, THREE SEEDS; fp16 ROWS WITH † HAVE FIVE). FAILED: BEST DEV SCORE MORE THAN 10 POINTS BELOW LoRA’S MEDIAN.

Method	fp16			fp32		
	Mean	Med.	Failed	Mean	Med.	Failed
Full fine-tuning	76.2 $\pm$ 1.9	75.5	0/3	77.6 $\pm$ 1.2	77.3	0/3
Linear probe	48.8 $\pm$ 2.0	48.5	–	49.4 $\pm$ 2.0	49.1	–
BitFit	78.2 $\pm$ 0.6	78.5	0/3	78.9 $\pm$ 1.4	79.5	0/3
LoRA †	78.0 $\pm$ 1.4	78.3	0/5	78.9 $\pm$ 0.8	79.3	0/3
DoRA	57.9 $\pm$ 37.8	79.1	1/3	78.4 $\pm$ 0.2	78.4	0/3
PiSSA	51.2 $\pm$ 33.0	61.5	2/3	78.3 $\pm$ 0.8	78.2	0/3
AdaLoRA	78.1 $\pm$ 0.7	77.7	0/3	78.9 $\pm$ 0.9	78.4	0/3
EVA †	59.4 $\pm$ 25.4	76.5	2/5	14.3 $\pm$ 0.0	14.3	3/3
EVA (whitened) †	77.3 $\pm$ 1.3	77.8	0/5	76.7 $\pm$ 0.6	76.9	0/3
DRIFT †	75.7 $\pm$ 8.0	79.0	1/5	79.3 $\pm$ 0.7	79.2	0/3
GEV	77.6 $\pm$ 2.2	78.3	0/3	56.7 $\pm$ 36.7	76.6	1/3

seed 1 stops at 61.4 F1 while its other four reach 78.7–80.0). LoRA, whitened EVA, AdaLoRA, BitFit and full fine-tuning never fail in Table I. For these methods the median over seeds, not the mean, describes a typical run: DoRA’s median is 79.1 and DRIFT’s 79.0, against LoRA’s 78.3. The fp16 failures are not reproducible run to run. DRIFT’s seed-1 tuning run and its seed-1 main-table run share every setting and differ only in GPU non-determinism, yet reach 82.5 and 66.2 dev F1.

Numerics or method? Rerunning the column in fp32 at the same rates (Table II) separates the two explanations. The four fp16 failures of PiSSA, DoRA and DRIFT disappear: none of their nine fp32 runs fails, including every seed that failed in fp16. These failures therefore belong to fp16 training near the edge of stability. EVA’s failures do not disappear. All three of its fp32 seeds, including the two that trained in fp16, settle on a constant prediction in the first epochs, with a training loss that stays at 0.38 from the second epoch and bounded gradients (maximum norm 2.8–8.3, against 143 in a PiSSA run that trained normally). EVA’s failure is thus an early collapse and not a divergence. It is a property of its initialisation

at its selected rate, and fp16 noise sometimes rescued runs from it rather than causing it. Nor is fp32 uniformly kinder: GEV, which fails no fp16 run, loses one of its three fp32 HoC seeds to the same early collapse (constant prediction from the first epoch, maximum gradient norm 1.7), and that seed was already its slowest in fp16. Across both precisions, the four initialisations line up with the energy of their initial directions: whitened EVA (1 $\times$ by construction) fails nothing, DRIFT (1.5–2.7 $\times$) one fp16 run and no fp32 run, GEV (8–15 $\times$) no fp16 run and one fp32 run, and EVA (59–73 $\times$) two of five in fp16 and all three in fp32. However, fp32 does not make training reproducible. A repeat of DRIFT’s fp32 seed-1 run, identical in every setting, reaches 79.5 against 78.8, because the memory-efficient attention backward stays non-deterministic in the mode we use (Section IV). If EVA’s HoC rate is selected by the mean dev score over three fp32 seeds, as the numerics objection asks, the selection picks 10^{-4} : there EVA trains in every seed but reaches 76.1 ± 1.2 , still 2.7 points below LoRA’s fp32 runs ($p = 0.007$). One step lower again, at 3×10^{-5} , it is merely undertrained (61.1 ± 2.0). One grid step lower in fp16 it neither fails nor gains: 77.4 ± 0.8 over five seeds, none failed, 0.6 points below LoRA ($p = 0.42$). Stabilising EVA, whether by whitening, by deflation or by a smaller step, never turns into accuracy.

Two robustness checks. The instance-level intervals of Appendix C come from a replication of every parameter-efficient run of Table I that stores its predictions, and they agree with the seed-level tests: of the 24 paired differences to LoRA only one excludes zero, BitFit’s loss on ChemProt (-1.1 , $[-2.0, -0.2]$). On HoC’s 371 test documents, the intervals of the methods that do not fail are ± 2 –3 points wide. Replacing RoBERTa’s 0.60M-parameter head with a single linear classification head (Appendix D) changes no conclusion either. No method beats LoRA significantly with this head (the largest difference is DRIFT’s $+0.8$, $p = 0.14$), and EVA remains the weakest (-1.5 , $p = 0.03$ uncorrected). The head that every method shares is therefore not what hides their differences.

Selection at the edge of stability. On a task this small, single-seed learning-rate selection lands near the edge of stability. This happens most often for methods that start from a data-derived subspace, consistent with the higher curvature Lee et al. [11] find for PiSSA, but not only for them. At AdaLoRA’s selected rate on RCT-20k, two of its three seeds fall to 44 and 61 dev F1 in the third epoch and only partly recover, and feed-forward-only LoRA at rank 14 fails two of three HoC seeds at its selected rate (Section V-D). AdaLoRA’s rate also shows how little separates these selections. On seed 1, its dev scores at 3×10^{-4} , 10^{-3} and 3×10^{-3} are 84.4, 84.5 and 84.5, and the rule takes the last. Averaged over three seeds, the dev score at the selected rate falls to 82.5 against 84.4 one grid step lower, where every seed trains and the test score is 84.0 ± 0.1 instead of 82.3 ± 1.8 . Wherever we checked the alternative (EVA on HoC in fp32 above, AdaLoRA here, and feed-forward-only rank 14 on HoC in Section V-D), selection by the mean over seeds picks the more stable rate. We keep single-seed selection everywhere, as the protocol states, and

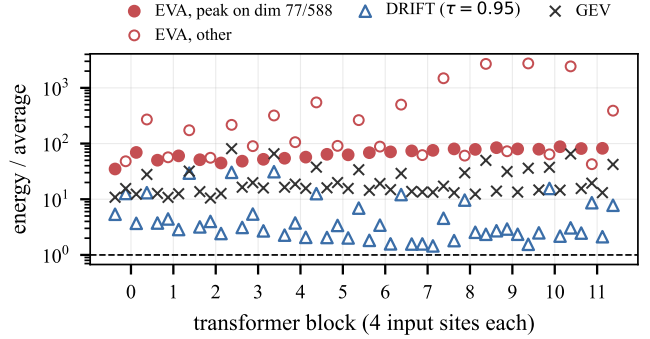


Fig. 1. Input energy along each module’s leading initial direction, relative to an average direction (log scale; RoBERTa-base, ChemProt). One point per input site: W_Q , W_K and W_V read the same block input. Filled: EVA’s leading direction peaks on outlier dimension 77 or 588; hollow circles: the attention-output and feed-forward hidden inputs, which are not in the hidden-state basis. Open triangles: DRIFT after deflation; crosses: GEV.

report the resulting runs as they are instead of re-selecting after seeing test scores.

EVA’s initialisation is the fragile part. Initialised along the target’s principal directions, EVA (budget-matched) trails LoRA on ChemProt with four times LoRA’s seed standard deviation (81.5 ± 1.2 vs. 82.5 ± 0.3 , five seeds) and falls to 59.4 ± 25.4 on HoC. Whitening its initial directions (Section IV) removes the HoC failures (77.3 ± 1.3 , none in five seeds). On ChemProt it lifts the mean to 82.0 ± 2.0 at its tuned rate of 10^{-3} , where one of five seeds starts slowly and ends at 78.5. At 3×10^{-4} , all five seeds reach 82.2 – 83.3 (82.7 ± 0.4). Whitening therefore moves the edge of stability up instead of removing it. This already suggests that the problem lies in the energy of the chosen directions, not in the rank allocation, and Section V-B identifies what those directions are.

B. RQ2: why activation-PCA initialisation is fragile

For most RoBERTa-base modules, the leading principal direction of the input covariance is one of the backbone’s two known outlier dimensions rather than a direction of biomedical meaning (Fig. 1). In the modules that read the hidden state (W_Q , W_K and W_V , which share the block input, and the feed-forward up-projection), the leading direction peaks on dimension 588 or 77, the outlier dimensions identified for RoBERTa-base in [36], [37]. This happens at all 24 distinct hidden-state inputs on ChemProt and at 23 of 24 on RCT-20k and HoC. Dimension 588 dominates on ChemProt, and dimension 77 on HoC. These directions carry a median 56–71 $\times$ the input energy of an average direction. The other inputs are no safer. The leading directions of the attention-output inputs carry a median 50–63 $\times$, and those of the feed-forward hidden activations a median 254–592 $\times$ and up to 2,748 $\times$, and over all 72 modules the median is 59–73 $\times$. (This ratio of the leading to the mean eigenvalue is the spectral counterpart of a small participation ratio, in which one direction holds a large share of the variance.) An adapter whose input-side factor points along such a direction sees input energy one to

three orders of magnitude larger than a randomly initialised one. A learning rate that is comfortable for LoRA therefore becomes a very large effective step: in the terms of [11], curvature rises and the usable learning rate falls. This matches several observations. First, EVA fails on HoC, where single-seed tuning selected LoRA’s rate for it (Section V-A). Second, it selects a rate one grid step below LoRA’s on ChemProt and RCT-20k (Appendix A). Third, the account explains its behaviour at low rank, where the fixed scale α/r is largest. At ranks 1 and 2 it collapses to the constant prediction already at 3×10^{-4} , whereas LoRA, whitened EVA and DRIFT still train at 10^{-3} , and at 10^{-4} its rank-1 adapter reaches 78.2 ± 2.0 with the best dev epoch the last one in every seed (a three-epoch probe at that rate had not yet left the constant prediction, so it trains late rather than never). Finally, EVA recovers under whitening, which shrinks these rows.

The general-domain reference carries the same outlier dimensions, which are a property of the backbone and not of biomedical text, so the reference subspace absorbs them and deflation removes them. Averaged over the 768-dimensional inputs, DRIFT’s leading drift direction puts a weight of 0.009–0.017 on dimension 588 (against 0.18–0.46 for EVA) and carries a median of only 1.5–2.7 $\times$ average energy. What the reference removes first is therefore not “general English” in a linguistic sense, as our motivating argument assumed, but the backbone’s own idiosyncratic high-energy dimensions.

The reference corpus matters only through the backbone.

To test how much of this is linguistic, we replace WikiText with news articles, with WikiText whose word order is shuffled within each passage, or with uniformly random tokens. In every case the reference subspace still absorbs the outlier dimensions. At $\tau = 0.95$ no leading drift direction peaks on dimension 77 or 588 at any of the 24 hidden-state inputs, and the leading directions carry a median 3–4 $\times$ the energy of an average direction with news or shuffled text and 13–15 $\times$ with random tokens, against 1.5–2.7 $\times$ with WikiText and 59–73 $\times$ without deflation. Accuracy follows none of these differences (Table III, three seeds). On ChemProt, DRIFT reaches 82.0, 83.1 and 82.8 with the news, shuffled and random-token references, against 82.3 with WikiText. On HoC it reaches 77.8 and 78.4 with news and random tokens, against WikiText’s 73.2, which includes DRIFT’s failed seed. No reference differs significantly from WikiText ($p \geq 0.07$). The shuffled reference’s gain over LoRA on ChemProt (+0.6, $p = 0.0007$ uncorrected) rested on three nearly identical paired differences, the case in which a t -test with two degrees of freedom is least reliable, so we ran seeds 4 and 5 of every reference control. Over five seeds the gain shrinks to +0.4 ($p = 0.05$ uncorrected; $p = 0.24$ after Holm correction over the five reference-versus-LoRA comparisons), the other references are within 0.2 points of LoRA, and none of the ten HoC runs with a replaced reference fails, against one of DRIFT’s five with WikiText. A reference made of random tokens is thus as good as real text: what the contrast contributes is the backbone’s outlier structure, which any input exposes, not knowledge of general English.

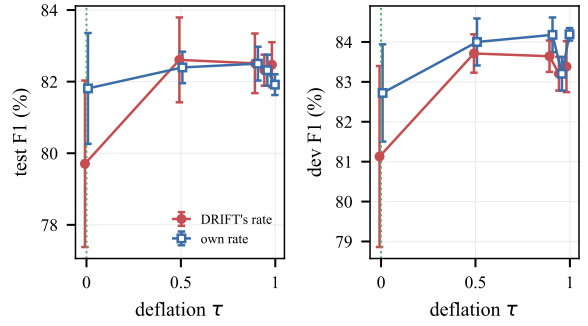


Fig. 2. Effect of the deflation level τ on ChemProt (mean $\pm$ s.d., three seeds; test F1 left, best dev F1 right), at DRIFT’s learning rate (filled) and at the rate selected for each level (open). At $\tau = 0$ nothing is deflated and DRIFT coincides with EVA. $\tau = 0.95$ was fixed before any experiment, so the sweep is descriptive.

The exact contrast gives the same outcome. The exact ratio criterion behaves like its two approximations (Section III-D). GEV’s leading directions avoid the outlier dimensions at all 24 hidden-state inputs and carry a median 8–15 $\times$ average energy (Fig. 1). Tuned on its own (10^{-3} , 10^{-4} and 3×10^{-4}), GEV reaches 82.4, 83.9 and 77.6 on ChemProt, RCT-20k and HoC. This is within 0.3 points of LoRA on every task ($p \geq 0.72$), with no failed run in fp16, although one of its three fp32 HoC seeds collapses (Section V-A), as its higher initial energy predicts. EVA’s own rank-unit rule, under which feed-forward rank costs no more of the budget than attention rank, spends 1.26M (ChemProt) and 1.29M (HoC) parameters instead of 1.33M and reaches 81.9 ± 0.5 and 78.4 ± 1.1 at EVA’s rate. On ChemProt this is level with budget-matched EVA (+0.1, $p = 0.87$). The rule fails none of its three HoC seeds, including seed 2, on which budget-matched EVA fails. Since fp16 failures are not reproducible from run to run, however, three seeds cannot show that the rule itself is more stable.

Removing the high-energy directions restores most of the stability but not accuracy. Without deflation ($\tau = 0$, i.e. EVA’s initialisation and allocation) and at DRIFT’s learning rate, the method reaches 79.7 ± 2.3 on ChemProt, whereas any $\tau \in [0.5, 0.99]$ gives 82.3–82.6 with a standard deviation of 0.4–1.2 (Fig. 2). At the rate selected for each level, which for $\tau = 0$ is EVA’s own, the gap shrinks to at most 0.7 points ($\tau = 0$: 81.8 ± 1.5 ; $\tau \geq 0.5$: 81.9 – 82.5 with standard deviations of 0.3–0.5). The robust effect is the collapse of the scatter rather than a gain in the mean. The curve is flat from $\tau = 0.5$ onward at either rate, so deflating even half of the reference energy removes what matters. Whitening achieves the same without a reference corpus, by rescaling the directions instead of removing them. Yet no member of the family carries accuracy past LoRA. Whitened EVA, DRIFT and GEV are within 0.5 points of LoRA on ChemProt and RCT-20k and below it on HoC (Tables I and III).

TABLE III

ABLATION (TEST F1, MEAN $\pm$ S.D.; SEEDS 1–3 THROUGHOUT, SO THE LoRA AND DRIFT ROWS DIFFER FROM THE FIVE-SEED VALUES OF TABLE I; 1.33M ADAPTER PARAMETERS EXCEPT EVA’S RANK-UNIT RULE, WHICH SPENDS 1.26–1.29M). OVER FIVE SEEDS THE REFERENCE ROWS GIVE ChemProt 82.3, 82.9, 82.6; HoC 78.0, 77.8 (SECTION V-B). UPPER BLOCK: ONLY THE ALLOCATION RULE AND THE INITIALISATION CHANGE, AT DRIFT’S LEARNING RATE. LOWER BLOCK: THE GENERALISED-EIGENVECTOR CONTRAST (OWN RATE), EVA WITH ITS OWN BUDGET RULE (EVA’S RATE), AND DRIFT WITH ITS WIKITEXT REFERENCE REPLACED. —: NOT RUN.

Variant	ChemProt	RCT-20k	HoC
LoRA (uniform rank, random init)	82.5 $\pm$ 0.3	83.9 $\pm$ 0.2	77.9 $\pm$ 1.3
Rank allocation only (random init)	82.1 $\pm$ 0.3	83.9 $\pm$ 0.1	65.0 $\pm$ 25.1
Drift init only (uniform rank)	82.9 $\pm$ 0.9	83.9 $\pm$ 0.2	78.7 $\pm$ 1.1
Random orthonormal init, drift ranks	82.3 $\pm$ 1.0	—	—
No deflation ($\tau=0$), DRIFT’s rate	79.7 $\pm$ 2.3	—	—
Absolute (unnormalised) drift score	82.4 $\pm$ 0.6	—	—
DRIFT (full)	82.3 $\pm$ 0.4	84.0 $\pm$ 0.1	73.2 $\pm$ 10.2
GEV init (uniform rank)	82.4 $\pm$ 0.6	83.9 $\pm$ 0.1	77.6 $\pm$ 2.2
EVA, rank-unit budget (its own rule)	81.9 $\pm$ 0.5	—	78.4 $\pm$ 1.1
DRIFT, news reference	82.0 $\pm$ 0.2	—	77.8 $\pm$ 1.9
DRIFT, word-shuffled reference	83.1 $\pm$ 0.3	—	—
DRIFT, random-token reference	82.8 $\pm$ 0.4	—	78.4 $\pm$ 0.8

C. RQ3: allocation or initialisation?

Table III factorises the design space. DRIFT changes two things at once, namely which modules get rank and which subspace each adapter starts in, so we vary the two independently on all three tasks. On ChemProt we also add a random-orthonormal initialisation, which separates the *choice* of subspace from the mere *scale* of an orthonormal initialisation. That scale differs from LoRA’s Kaiming-uniform default and would otherwise be a confound.

Allocation does not help, and initialisation is neutral to positive. Combining DRIFT’s rank allocation with LoRA’s random initialisation scores lower than uniform LoRA in every seed on ChemProt (-0.3 , $p = 0.03$ uncorrected; like the other nominal results, suggestive only). It is level on RCT-20k ($+0.03$) and destabilising on HoC, where one of its three seeds fails (65.0 ± 25.1). DRIFT’s initialisation with uniform ranks is the best low-rank configuration in the upper block on ChemProt (82.9 ± 0.9), though not significantly better than LoRA ($p = 0.46$). On RCT-20k and HoC it is level with LoRA ($+0.04$ and $+0.8$, $p \geq 0.41$), with no failed run. Replacing the drift subspace with a random orthonormal one under the same ranks leaves the mean unchanged (82.3) and doubles the spread, so the orthonormal scale is not a confound. The unnormalised score changes nothing either. Allocation is thus the half of the method that costs stability, while initialisation is the half that is free.

Why allocation has little to work with. The drift ratio $\rho_m = \text{tr}(\tilde{\Sigma}_m)/\text{tr}(\Sigma_m^D)$ is small, and most of it is set by τ itself. If target and reference shared their geometry exactly, a fraction $1 - \tau$ of the target variance would fall outside the reference subspace by construction. Averaged over the 48 distinct input sites, the excess over $1 - \tau$ is only 2.5 (ChemProt), 2.2 (RCT-20k) and 1.4 (HoC) points of variance at $\tau = 0.95$ ($\rho_m = 7.5$, 7.2 and 6.4%), and 6.4/5.0/4.9 and 0.75/0.56/0.37 points at

TABLE IV

ADAPTER PLACEMENT WITH LoRA, EACH CONFIGURATION AT ITS OWN TUNED LEARNING RATE (TEST F1, MEAN $\pm$ S.D., SEEDS 1–3; TABLE I REPORTS FIVE SEEDS FOR THE METHODS THAT HAVE THEM). RANK 14 ON THE FEED-FORWARD MATRICES SPENDS ABOUT THE ALL-MODULE RANK-8 BUDGET; RANK 4 ON ALL MODULES ABOUT THE FEED-FORWARD RANK-8 BUDGET. DRIFT ROWS USE DRIFT’S RATE. TWO OF THREE SEEDS FAIL IN THE FEED-FORWARD RANK-14 HoC CELL AT ITS SELECTED RATE; AT THE RATE A THREE-SEED DEV MEAN SELECTS, ALL THREE TRAIN AND IT REACHES 77.0 $\pm$ 4.1 (SECTION V-D).

Placement	r	Params	ChemProt	RCT-20k	HoC
All modules	8	1.33M	82.5 $\pm$ 0.3	83.9 $\pm$ 0.2	77.9 $\pm$ 1.3
All modules	4	0.66M	82.6 $\pm$ 0.5	83.5 $\pm$ 0.5	78.7 $\pm$ 1.0
Feed-forward only	8	0.74M	83.4 $\pm$ 0.9	83.7 $\pm$ 0.2	78.0 $\pm$ 1.4
Feed-forward only	14	1.29M	82.8 $\pm$ 0.8	83.6 $\pm$ 0.4	35.0 $\pm$ 35.9
Attention only	8	0.59M	82.3 $\pm$ 0.5	83.9 $\pm$ 0.1	77.9 $\pm$ 3.6
DRIFT, feed-forward only	8	0.74M	82.8 $\pm$ 0.3	83.8 $\pm$ 0.3	75.6 $\pm$ 7.8
DRIFT, attention only	8	0.59M	81.2 $\pm$ 0.8	83.7 $\pm$ 0.1	78.2 $\pm$ 1.5

$\tau = 0.5$ and $\tau = 0.99$. In other words, biomedical text largely reuses the geometry of general English. The ordering is the same at every τ and matches lexical distance from general text, with ChemProt, dense with chemical and protein names, drifting most. Drift is highest in the first four blocks and roughly flat beyond them, which is consistent with domain shift entering as unfamiliar terminology at the lexical end of the network. Because the differences between sites and tasks amount to a few points of variance, allocation driven by them has little to exploit. A divergence that does not depend on τ tells the same story (Appendix E). In CORAL distance, each task’s input covariances sit at 10–15 times the floor between two disjoint halves of WikiText (1.17 for ChemProt, 1.10 for the clinical notes of Section V-F, 0.88 for RCT-20k, 0.76 for HoC, against 0.08). They follow the same order as the drift ratio, and the Bures and log-det divergences agree. The scale reflects what a corpus is rather than how useful it is as a reference: news text sits closer to WikiText (0.92) and uniformly random tokens further away (1.06) than two of the tasks, yet both work equally well as references (Section V-B).

The resulting allocations differ visibly. Under a *parameter* budget, an attention rank unit costs $d^{\text{in}} + d^{\text{out}} = 1,536$ parameters against 3,840 for a feed-forward one, so both methods buy cheap attention capacity first. This is a property of budgeting rather than of either criterion. Where the two methods differ is in depth. DRIFT holds the attention projections of the first five blocks at the cap (38% of modules at the cap, against 18% for EVA) and withdraws rank from blocks 7–10, driving their feed-forward down-projections to zero. The fact that this visibly different allocation yields the same accuracy as uniform rank (Table III) is the clearest evidence that, at this budget, allocation is not where the accuracy lies.

D. RQ3: where should the adapter go?

Allocation moves rank between modules that all carry an adapter, whereas placement decides which modules carry one at all. Adapting only the feed-forward matrices at rank 8 spends 0.74M parameters instead of 1.33M, so a comparison with all-module rank 8 confounds placement with budget. This

is the introduction’s warning that a matched rank is not a matched budget, in reverse. The placement study (Table IV) therefore compares each placement with the other at its budget, on all three tasks, with every configuration tuned on its own. The pairs are feed-forward-only rank 14 (1.29M) against all-module rank 8 (1.33M), and all-module rank 4 (0.66M) against feed-forward-only rank 8 (0.74M). The second pair also tests the alternative explanation that feed-forward-only LoRA wins merely by having fewer parameters to fit on small data.

At matched budgets. Neither pair shows a reliable difference. Against all-module rank 8, feed-forward-only rank 14 is +0.3 on ChemProt ($p = 0.52$) and -0.2 on RCT-20k ($p = 0.53$). On HoC, two of its three seeds never leave the constant prediction at its selected rate of 10^{-3} , so that cell settles nothing until the rate is chosen differently (next paragraph). Against feed-forward-only rank 8, all-module rank 4 is -0.8 on ChemProt ($p = 0.10$), -0.2 on RCT-20k ($p = 0.65$) and $+0.7$ on HoC ($p = 0.07$). The lead of feed-forward-only rank 8 over all-module rank 8 on ChemProt ($+0.9$, $p = 0.32$) is therefore not a placement effect that survives budget matching. Nor are fewer parameters an advantage: all-module rank 4 (0.66M) differs from all-module rank 8 by $+0.1$, -0.4 and $+0.8$ points ($p \geq 0.34$), and attention-only rank 8 (0.59M) by at most 0.2. At this budget and below it, capacity is not what limits accuracy, and a larger budget helps only on HoC (Section V-E).

Repairing the uninformative HoC cell. A cell in which two of three seeds never train supports no comparison at all, so we asked separately what causes the failures and what a stricter rule would have selected. Precision explains part of it, but not all. When rerun in fp32 at the same rate, one of the two failed seeds recovers (77.3 and 78.1 against a third still at 14.3). This places the configuration between the fp16 failures that fp32 removes completely and EVA’s, which it does not remove at all (Section V-A). The selection rule accounts for the rest. Averaged over the three seeds, the dev score is 35.3 at the selected 10^{-3} and 80.7 at 3×10^{-4} , so choosing the rate by the multi-seed mean takes the lower one, where all three seeds train and the test score is 77.0 ± 4.1 . The upward budget-matched comparison therefore rests on that rate rather than on fp32, and it reaches the same place. Feed-forward-only rank 14 is 0.9 points below all-module rank 8 ($p = 0.78$) and 1.0 below feed-forward-only rank 8. Placement buys nothing at a matched budget on HoC either. What the cell adds is a third case, after EVA and AdaLoRA, in which single-seed selection lands on a rate the configuration cannot reliably train at.

Relation to prior work. He et al. [13] found that feed-forward modification uses added parameters more effectively than attention modification, on larger backbones and mostly on generation tasks. We do not find this with matched budgets, separately tuned learning rates and small biomedical encoders. Feed-forward-only LoRA is never reliably better than all-module LoRA at the same budget, and at rank 14 it is the least stable configuration on HoC, where it needs a multi-seed learning rate to train at all. Adapting only attention matches all-module LoRA on every task with less than half its parameters, and DRIFT placed on one module type is

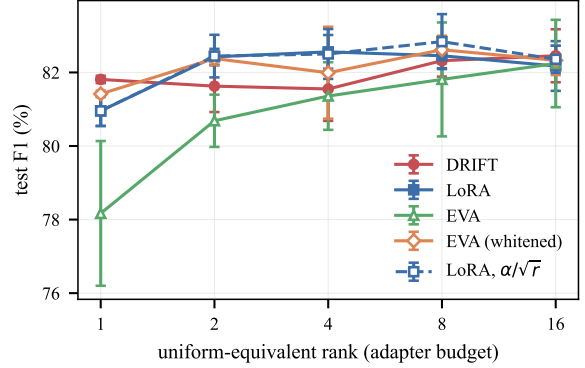


Fig. 3. Test micro-F1 against adapter budget (uniform-equivalent rank) on ChemProt, mean $\pm$ s.d. over three seeds, every (method, budget) pair at its own tuned learning rate (Appendix A).

TABLE V
BACKBONE LADDER ON CHEMPROT (TEST MICRO-F1, MEAN $\pm$ S.D., THREE SEEDS):
THE BERT MINIATURES SHARE ONE VOCABULARY AND PRETRAINING RECIPE.
EACH METHOD RUNS AT THE LEARNING RATE TUNED ON THAT BACKBONE’S DEV
SET; THE LAST COLUMN KEEPS EVA AT THE RATE TUNED FOR IT ON
RoBERTA-BASE.

BERT	LoRA	EVA	EVA (whitened)	DRIFT	EVA (RoBERTa’s rate)
Tiny (4.4M)	60.9 $\pm$ 1.9	58.3 $\pm$ 1.1	59.3 $\pm$ 1.2	61.2$\pm$1.8	47.5 $\pm$ 3.4
Mini (11.2M)	70.8$\pm$0.9	68.4 $\pm$ 2.1	69.1 $\pm$ 0.8	70.1 $\pm$ 0.9	63.9 $\pm$ 0.8
Small (28.8M)	76.1$\pm$0.7	74.4 $\pm$ 0.4	74.6 $\pm$ 0.4	74.8 $\pm$ 0.7	71.1 $\pm$ 0.7
Medium (41.4M)	77.6$\pm$0.7	76.5 $\pm$ 1.3	76.7 $\pm$ 0.5	76.6 $\pm$ 1.1	74.9 $\pm$ 0.9
Base (110.1M)	79.2 $\pm$ 0.9	79.2 $\pm$ 1.3	79.8 $\pm$ 0.4	80.5$\pm$0.5	79.2 $\pm$ 1.3

never significantly better than LoRA placed there (it trails on ChemProt with attention only, -1.1 , $p = 0.23$, and fails one HoC seed with feed-forward only). Reallocating the budget between modules, which is what every allocation method optimises, moves accuracy no more than this coarse choice does, and at this budget neither moves it measurably.

E. RQ4: budget, scale and architecture

Budget. Across budgets from rank 1 to 16 on ChemProt (Fig. 3), with the learning rate tuned at every budget, LoRA, DRIFT and whitened EVA stay within 1.0 points of each other at every budget, and LoRA is flat from rank 2 (82.4) to rank 16. EVA trails at the smallest budgets (-2.8 and -1.7 against LoRA at ranks 1 and 2; $p = 0.10$ and 0.03 uncorrected) and catches up by rank 16. Tuning keeps its rate at 10^{-4} up to rank 8, half a decade to a decade below the other methods’ rates. Where the budget binds hardest, at rank 1, DRIFT has the best mean (81.8 ± 0.1 against LoRA’s 81.0 ± 0.4 ; $+0.9$, $p = 0.04$ uncorrected, not significant after Holm correction over the sweep’s twelve comparisons with LoRA). This is the one place in our study where allocation might have room to help. We keep the scale α/r , as EVA does, rather than rsLoRA’s $\alpha/\sqrt{r}$ [26]. Under Adam, changing the scale multiplies every update of sBA by the same factor, and per-rank tuning of the learning rate absorbs this to first order. The dashed curve in Fig. 3 tests this argument. LoRA with $\alpha/\sqrt{r}$, tuned at every

rank, stays within 0.4 points of LoRA with α/r at every budget (largest difference +0.4 at rank 8, $p = 0.58$), and at rank 1, where the two scales coincide, the two runs reproduce each other exactly. Per-rank tuning thus absorbs the change of scale.

Budget on the other tasks. Whether the ranking changes with the budget is not only a question for ChemProt. At 0.5% (rank 4) and 2% (rank 16) on RCT-20k and HoC, with every method and budget tuned on its own (Appendix F), the results match the picture that Table I gives at rank 8. On RCT-20k the four methods lie within 0.6 points of each other at every budget, with whitened EVA and DRIFT marginally above LoRA at rank 4 (+0.4, $p \geq 0.25$), while on HoC LoRA has the best mean at all three budgets. There, EVA trails it by 1.4 at rank 4 and 2.5 at rank 16, whitened EVA by 1.3 at both, and DRIFT loses a seed at rank 4 (61.3 ± 29.1) and trails by 5.4 at rank 16. Two differences reach $p < 0.05$ uncorrected, both of them losses to LoRA (EVA at rank 16 on RCT-20k, -0.5 ; DRIFT at rank 16 on HoC, -5.4), and neither survives Holm correction over the table’s twelve comparisons. HoC is also the one place where the budget itself matters. LoRA at rank 16 beats LoRA at rank 8 by 2.4 points ($p = 0.05$ uncorrected), while rank 4 is level with rank 8 on all three tasks. So where a larger budget helps, allocating it differently still does not.

Model scale. Table V tests the prediction from Section I that allocation should matter more as the backbone shrinks, now with every method tuned on every backbone. The prediction is not borne out. LoRA has the best mean on BERT-Mini, -Small and -Medium, and DRIFT on BERT-Tiny (+0.3, $p = 0.89$) and BERT-Base (+1.3, $p = 0.03$ uncorrected). The latter is the largest gain over LoRA in the ladder but not significant after Holm correction over its twenty comparisons with LoRA. At their own tuned rates, EVA trails LoRA by 0.0 to 2.5 points and whitened EVA by at most 1.8, none significantly. The last column keeps EVA at the rate tuned for it on RoBERTa-base (10^{-4}), as our first version of this study did. EVA then falls further behind as the backbone shrinks: it is level on BERT-Base but 2.7, 4.9, 6.9 and 13.4 points behind the tuned LoRA from BERT-Medium down to BERT-Tiny. The one difference in the ladder that survives correction is this column’s loss on BERT-Mini. Most of that deficit was caused by the inherited learning rate and not by the initialisation, which is the clearest single illustration of why a rate tuned for one backbone is not a neutral choice for another. After per-backbone tuning, what remains is a small EVA deficit that grows as the backbone shrinks (0.0 on BERT-Base, 1.2 on BERT-Medium, 1.7 on BERT-Small, 2.5 on BERT-Mini and BERT-Tiny).

Profile noise on small backbones. A narrow backbone profiled from 1,024 sequences might simply get noisy directions. We therefore profiled BERT-Tiny from all 4,169 ChemProt training sentences and all 3,227 WikiText passages (four and three times the standard profile, and all the text these corpora have). This changes EVA, whitened EVA and DRIFT by +0.7, +1.3 and -0.9 points, none significantly ($p \geq 0.20$). EVA remains 1.8 points behind LoRA, so its shortfall on the smallest backbone is not mainly estimation noise in the profile.

Outlier dimensions across the ladder. Profiling the BERT

TABLE VI
DECODER SLM: SmoLLM2-360M (TEST F1, MEAN $\pm$ S.D., THREE SEEDS), EACH METHOD AT ITS OWN TUNED LEARNING RATE (RATE ABOVE THE SCORE).

Method	ChemProt	HoC
	10^{-3}	10^{-3}
LoRA	83.9 ± 0.5 3×10^{-4}	80.9 ± 0.8 10^{-3}
EVA (budget-matched)	82.2 ± 3.6 10^{-3}	74.6 ± 5.5 10^{-3}
EVA (whitened)	83.1 ± 1.4 10^{-3}	80.5 ± 0.3 10^{-3}
DRIFT	83.4 ± 1.9	80.3 ± 0.9

miniatures shows that the mechanism of Section V-B is not specific to RoBERTa, but also that it does not explain EVA’s deficit. If we count distinct inputs to the hidden-state modules (W_Q , W_K and W_V share one), a single dimension hosts EVA’s leading direction at 13 of 24 inputs on BERT-Base, in every layer, and at 11 of 16 on BERT-Medium, compared with 23 of 24 on RoBERTa-base. The pattern weakens to 5 of 8 on BERT-Mini and 3 of 8 on BERT-Small, and no dimension recurs across BERT-Tiny’s four inputs. The energy excess shrinks as well. EVA’s leading directions carry a median $59\times$ the energy of an average direction on BERT-Base, then $40\times$, $36\times$, $20\times$ and $11\times$ down to BERT-Tiny, because their share of the variance stays near 7–8% at the hidden-state inputs while the width falls from 768 to 128 (deflation cuts them to 1.6–4.9 $\times$). On its own, the mechanism therefore predicts milder instability on small backbones, not stronger, and the ladder agrees. At its tuned rate EVA fails no run on any BERT miniature, and its seed standard deviation (0.4–2.1 points) is of the order of LoRA’s (0.7–1.9). EVA’s small deficit, which grows as the backbone shrinks while the energy excess falls, must therefore have another cause, which these experiments do not isolate.

A decoder SLM. SmoLLM2-360M, a decoder-only SLM, reproduces the pattern seen on RoBERTa on ChemProt (Table VI), with every method tuned on its own dev set under the same protocol. Its outlier signature is the strongest we measured. Dimension 40 hosts EVA’s leading direction at 41 of the 64 hidden-state inputs, in layers 1–25 of 32. This is consistent with the massive activations reported for LLMs, which also sit in fixed dimensions of the middle layers [42]. EVA’s leading directions carry a median $125\times$ the energy of an average direction (6.3 $\times$ after deflation). Tuning again places EVA’s learning rate one grid step below LoRA’s (3×10^{-4} against 10^{-3} , just as 10^{-4} against 3×10^{-4} on RoBERTa-base’s ChemProt). Whitened EVA and DRIFT, whose initial directions carry little excess energy, select LoRA’s rate or a higher one on ChemProt on both backbones. This is what the curvature account of Section V-B predicts, with a lower usable rate for EVA alone. No method beats LoRA (83.9 ± 0.5), and EVA is again the least stable (82.2 ± 3.6 , one seed at 78.1). On HoC, where the encoder’s failures occurred and 512-token documents require a batch of 8, all four methods train without a failed run and tuning gives all of them 10^{-3} . LoRA reaches 80.9 ± 0.8 , whitened EVA 80.5 ± 0.3 and DRIFT 80.3 ± 0.9 ,

TABLE VII

CLINICAL NOTES (MTSAMPLER, 12 SPECIALTIES; TEST MICRO- AND MACRO-F1, MEAN $\pm$ S.D. OVER THREE SEEDS). EACH METHOD RUNS AT THE RATE TUNED ON THE CLINICAL DEV SET; THE REFERENCE ROWS USE DRIFT’S RATE. Δ : PAIRED DIFFERENCE TO LoRA IN MICRO-F1. FAILED: AS IN TABLE I.

Method	Micro-F1	Macro-F1	Δ (p)	Failed
LoRA	84.6 $\pm$ _{1.0}	84.3 $\pm$ _{1.8}	–	0/3
EVA (budget-matched)	83.5 $\pm$ _{0.2}	81.5 $\pm$ _{1.2}	-1.1 (0.20)	0/3
EVA (whitened)	83.4 $\pm$ _{0.6}	82.7 $\pm$ _{1.0}	-1.2 (0.13)	0/3
DRIFT	84.5 $\pm$ _{0.2}	83.4 $\pm$ _{0.8}	-0.1 (0.86)	0/3
DRIFT, news reference	84.4 $\pm$ _{1.4}	83.3 $\pm$ _{1.2}	-0.2 (0.74)	0/3
DRIFT, random-token reference	85.1 $\pm$ _{0.6}	84.4 $\pm$ _{0.4}	+0.4 (0.66)	0/3

while EVA is again the least stable at 74.6 ± 5.5 , with seeds at 80.6, 73.3 and 69.9 (-6.3 against LoRA, $p = 0.22$). In fp16 the decoder thus repeats the encoder’s pattern, with the same ordering and instability concentrated in EVA, but the cause differs. In two of EVA’s three seeds the per-epoch logs show the fp16 loss scale collapsing to zero, with no gradient reaching the adapter from the fifth epoch on, so their scores come from checkpoints of the second and third epochs, which stay above our failure threshold. Rerun in fp32 (Section IV) at the same rate and from the same profile, EVA trains in every seed without a non-finite step, although its gradient norm before clipping still reaches 291, against at most 12.8 for LoRA. It scores 79.8 ± 1.7 , level with LoRA’s 80.7 ± 1.5 in fp32 (-0.9 , $p = 0.66$), and precision leaves LoRA itself unchanged (-0.2 , $p = 0.91$). On the decoder, EVA’s instability is fp16 overflow, not the early collapse it shows on RoBERTa-base, and removing it again adds no accuracy.

F. Clinical notes

PubMed abstracts are stylistically close to WikiText, but clinical notes, with their abbreviations, templated sections and dictation style, are not. To test whether our findings survive the shift our deployment motivation implies, we add a clinical task: classifying transcribed clinical notes from MTSamples [57] by medical specialty. The public release lists many notes under both a specialty and a document type (e.g. “Surgery” or “Discharge Summary”). We drop the document-type and catch-all labels, drop the notes that remain listed under two specialties, and keep the 12 specialties with at least 70 notes (1,972 notes), split 70/15/15 within each specialty and truncated to 512 tokens. LoRA, EVA, whitened EVA and DRIFT are each tuned on the clinical dev set, and DRIFT also runs with the news and random-token references (Table VII).

Both the mechanism and the conclusions carry over. EVA’s leading directions peak on dimension 77 or 588 at 21 of the 24 hidden-state inputs of the clinical profile and carry a median $57\times$ the energy of an average direction, compared with 24 of 24 and $73\times$ on ChemProt. Deflation leaves one such input and a median of $4.0\times$. The drift ratio matches ChemProt’s, with an excess over $1 - \tau$ of 2.6 points at $\tau = 0.95$, against 2.5. Tuning selects 10^{-4} for all four methods, and no run fails. LoRA is again unbeaten—84.6 micro-F1 (84.3 macro) against DRIFT’s 84.5 (-0.1 , $p = 0.86$), EVA’s 83.5 (-1.1 ,

$p = 0.20$) and whitened EVA’s 83.4 (-1.2 , $p = 0.13$). The random-token reference gives the best mean in the table (85.1, $+0.4$ over LoRA, $p = 0.66$) and the news reference 84.4, so the reference corpus does not matter on clinical text either. Clinical style changes none of the quantities we measured, and the backbone’s outliers dominate its activations as they do on PubMed abstracts.

G. Cost of profiling

Profiling consists of two forward passes, one eigendecomposition of each reference covariance Σ_m^G , and one eigendecomposition of each projected target covariance per value of τ . On one NVIDIA T4, the GPU used for every training run, the two forward passes over 1,024 reference and 1,024 target sequences take 28–35 s at the short sequence lengths of ChemProt and RCT-20k and 99 s at HoC’s 512 tokens. The spectral step takes 45–47 s for one τ and 136–143 s for five. This is nearly independent of the task, because the cost is set by the module dimensions (768 and 3,072) rather than by the amount of text. The step runs on the CPU in double precision and could be moved to the GPU. In total, profiling at one τ costs 42–47% of a single LoRA training run on the same task and GPU. That is cheap, but not negligible, and it is not free relative to the accuracy it buys at this budget. The reference pass can be reused, since Σ_m^G depends only on the backbone, and a deployment that adapts one model to many domains pays for it only once. Because the adapter merges into the base weights after training, the inference cost is identical to that of LoRA and of the unadapted backbone.

VI. DISCUSSION: EVALUATING ADAPTER ALLOCATION

Inference cost does not separate the methods we compare. Every low-rank adapter merges into the backbone after training, so the costs that differ are training and profiling time, and the only question is whether profiling buys accuracy. Our results do not say that adapter capacity is irrelevant, since at rank 1 it does bind. What they say is that, for small encoders at a budget of about 1%, where the capacity goes (across modules or across module types) matters less than how each configuration is trained, that is, the learning rate selected for it and the energy of its initial directions. Fig. 4 condenses these lessons into a protocol, and we explain the reasons below.

Match parameters, and tune every configuration. A rank unit costs 1,536 parameters in an attention projection of RoBERTa-base and 3,840 in a feed-forward matrix, so equal total rank is not an equal budget. AdaLoRA, pruned to the total rank of uniform rank 8, ends at up to 1.43M parameters rather than 1.33M. Methods also favour different learning rates [11], and so do backbones and budgets. Without these controls, a new allocation rule can gain simply by spending more parameters, or by being compared at a learning rate chosen for another configuration. At the rate selected on RoBERTa-base, EVA fell 13.4 points behind the tuned LoRA on BERT-Tiny, and at the rate tuned on BERT-Tiny itself only 2.5.

Report spread and failures, not only means. On HoC the methods are separated by failed runs rather than by accuracy,

A protocol for evaluating adapter allocation

- 1) *Budget in parameters.* Match adapter parameters, not total rank, and report the parameters actually spent (head excluded).
- 2) *Tune every configuration.* Select the learning rate on the dev set separately for every method, backbone, budget and placement, and extend each grid until the selection is interior.
- 3) *Seeds and failures.* Use at least five seeds on small tasks. Report medians and failed runs next to means, rerun failures in fp32 to separate numerics from method, and correct for multiple comparisons.
- 4) *Energy check.* Before training a data-derived initialisation, compute each initial direction’s energy relative to an average direction (target forward pass plus one leading eigenvector per module). On RoBERTa-base and SmoLM2-360M, medians of 59–125× coincided with failed or unstable runs and 1–15× with stable training; on the BERT ladder 20–59× trained stably at tuned rates. Read a high median as a warning to lower the rate, whiten or deflate—not as a verdict.
- 5) *Check that the budget binds.* Before trying allocation, run LoRA at half the budget and at other placements of the same budget. If nothing changes, allocation has nothing to exploit. Neither check needs profiling.

Fig. 4. A protocol for evaluating adapter allocation, distilled from Section V.

and a single run can land on either side. Two fp16 runs of DRIFT with identical settings reached 82.5 and 66.2 dev F1. In fp32, the failures of PiSSA, DoRA and DRIFT vanished while EVA’s became universal, so the same count of failed runs can mean numerical noise for one method and a genuine instability for another. One method can even fall on both sides: EVA’s instability on the decoder vanished in fp32 (Section V-E). Medians, failure counts, an fp32 check and a multiplicity correction make such differences visible, instead of averaging them into a mean that describes no run.

Check the energy of a data-derived initialisation. On a backbone with outlier dimensions, these dimensions dominate the principal directions of the activations. Computing the energy of each initial direction relative to an average one (Fig. 1) costs only the target half of the profiling pass (10–63 s here) and a leading eigenvector per module. EVA’s medians were 59–73× on RoBERTa-base, where it failed runs, and 125× on SmoLM2-360M, where it failed no run but was the least stable method in fp16, against 1–15× for the initialisations that largely restored stability (whitened EVA, 1× by construction; DRIFT with any of our four references; GEV). The residual risk follows the same scale. GEV, the highest of these at 8–15×, is the one that still lost a run. The check is a warning rather than a verdict. On the BERT ladder, initialisations at 20–59× trained stably at tuned rates, including EVA’s 59× on BERT-Base. Recurring outlier dimensions also appear in BERT-Base, BERT-Medium and SmoLM2-360M, so the diagnostic is not specific to RoBERTa, but it is not decisive on its own either.

Check whether the budget binds. An allocation rule can only help where capacity is scarce, and at 1% of RoBERTa-base it usually is not. All-module LoRA at half the budget and

attention-only LoRA at less than half match the full budget on all three tasks, and on ChemProt rank 2, a quarter of it, is within 0.1 points. Where the budget does bind, the answer is the same. At rank 1 on ChemProt, DRIFT has its only nominal gain of the budget sweep, and on HoC, where doubling the budget to 2% gains LoRA 2.4 points, no allocation rule beats LoRA at that budget either. A budget sweep and a placement comparison need no profiling, whereas profiling costs almost half a training run (Section V-G).

What this means for defaults and for their authors. PEFT libraries ship the choices our baseline uses (adapters on all linear maps, $s = \alpha/r$, Kaiming-uniform A and $B = 0$), and our results support them as defaults for small encoders, with two amendments. First, report the parameters an adapter actually spends, since placement and allocation change them far more than they change accuracy here. Second, make data-derived initialisations report the energy of their directions. For EVA, the configuration we test differs from the published one in its budget unit, so Section V-B also reports the published rank-unit rule. That rule spends slightly fewer parameters, matches budget-matched EVA’s accuracy and failed none of its three HoC seeds, so our conclusions about EVA do not hinge on the budget unit. Our decoder results use last-token classification, and practitioners who use decoders generatively face a different objective, about which we make no claim.

Deployment. Everything here runs on a single commodity GPU with a frozen, shared backbone, which keeps sensitive data inside an institution. If anything, the move towards small on-premise models that this study supports is privacy-positive.

VII. LIMITATIONS

Input-side profiling cannot separate W_Q , W_K and W_V . These three projections consume the same block input, so they share a covariance matrix and receive identical drift scores. As a result, DRIFT resolves four distinct sites per block rather than six. Distinguishing them requires an output-side signal, which costs a backward pass and would give up the gradient-free property. Our allocation routine already accepts an optional per-module weight for exactly this purpose, and we leave a hybrid variant to future work.

The reference corpus is a proxy. Assumption (A) is stated with respect to the pretraining distribution, which is not public for most checkpoints. We substitute WikiText-103, a general-domain corpus that is clearly not RoBERTa’s training set, so (A) holds only approximately. We tested three substitutes for it (Section V-B), and each, random tokens included, served as well. We did not test references closer to the pretraining mix.

Statistical power and selection. Three to five seeds detect only large differences. Our conclusion that no method beats LoRA therefore rules out gains of the size typically claimed for adaptive allocation (a point or more), but not small ones. The runs behind Table I did not store per-example predictions, so its intervals come from a replication of every parameter-efficient run in it (Appendix C), and Appendix B lists every seed. Learning rates are selected on one seed, which on HoC puts several methods near the edge of stability.

Selecting EVA’s rate by the mean over three fp32 seeds instead (Section V-A) removes its failures but leaves it 2.7 points below LoRA. The controls of Table III inherit their parent configuration’s rate (Section IV). Differences inside the table’s upper block, the -0.3 of allocation alone among them, therefore compare variants at DRIFT’s rate rather than at each variant’s own, and are suggestive for that reason too. We did not re-select after observing test results.

Clinical text and first-order shift. Our motivation is clinical deployment, but apart from the MTSamples task our data are public PubMed text, which is stylistically much closer to WikiText than clinical notes are. The MTSamples notes are sample reports published as reference and training material for medical transcriptionists, and they are not records from an electronic health record. They carry no de-identification tokens and little of the noise of real notes, so they test clinical style rather than clinical deployment. All profile-based methods here also centre their inputs, so a mean shift between domains, which is plausibly a large part of the shift towards clinical text, is invisible to them by construction.

Scope. We study the classification of biomedical and clinical text. The full budget sweep, the τ sweep and the ladder use ChemProt only. The placement study and the allocation/initialisation factorisation cover all three benchmarks, the reference controls and the decoder cover ChemProt and HoC, the two further budgets (0.5% and 2%) cover RCT-20k and HoC, and the clinical task covers only the four methods on which the analysis turns. Architectures without pronounced outlier dimensions may not show EVA’s instability at all. Larger decoders, generative formulations and other specialised domains remain untested. HoC has only 371 test documents, so its confidence intervals are wide (Appendix C).

One profiling pass per task. The drift profile depends on the target corpus, so a new domain requires a fresh (cheap) profiling pass. Where many tasks share one domain, the profile can be reused, but we have not measured how much this loses.

VIII. CONCLUSION

Activation-guided allocation asks where the target distribution’s variance is largest. On RoBERTa-base, for most modules, the answer is one of two outlier dimensions of the backbone, and adapters initialised there are fragile. Whiten-ing those directions, deflating them with a general-domain reference (DRIFT), or taking the generalised eigenvectors of the target and reference covariances restores most of the stability without labels, gradients or inference cost. But under a matched parameter budget, with every configuration tuned on its own, no such fix and no allocation rule we tested outperformed uniform LoRA on three biomedical benchmarks, a clinical task or a decoder SLM. Rank allocation alone slightly hurt, adapter placement changed nothing reliable at a matched budget, and half the budget was enough. On smaller backbones the learning rate tuned for each backbone, not any allocation rule, decided how far EVA trailed LoRA.

We believe this is the right baseline for the field to report against. Improvements claimed for activation-aware allocation

should be measured at matched parameters, against a separately tuned LoRA, at learning rates tuned for the backbone, budget and placement in question, over enough seeds to reveal their failures, and checked for the stability of their initialisation. Natural next steps are to test the outlier mechanism in larger decoders, generative tasks and real clinical records, and to ask whether a cheap output-side signal can make allocation matter where input geometry alone does not.

REFERENCES

- [1] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [2] Z. Han, C. Gao, J. Liu, J. Zhang, and S. Q. Zhang, “Parameter-efficient fine-tuning for large models: A comprehensive survey,” *Transactions on Machine Learning Research*, 2024.
- [3] Q. Zhang, M. Chen, A. Bukharin, P. He, Y. Cheng, W. Chen, and T. Zhao, “AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [4] N. Ding, X. Lv, Q. Wang, Y. Chen, B. Zhou, Z. Liu, and M. Sun, “Sparse low-rank adaptation of pre-trained language models,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [5] M. Valipour, M. Rezagholizadeh, I. Kobayev, and A. Ghodsi, “DyLoRA: Parameter-efficient tuning of pre-trained models using dynamic search-free low-rank adaptation,” in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2023.
- [6] F. Paischer, L. Hauenberger, T. Schmied, B. Alkin, M. P. Deisenroth, and S. Hochreiter, “Parameter efficient fine-tuning via explained variance adaptation,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025, arXiv:2410.07170.
- [7] Y. Yang, X. Li, Z. Zhou, S. L. Song, J. Wu, L. Nie, and B. Ghanem, “CorDA: Context-oriented decomposition adaptation of large language models for task-aware parameter-efficient fine-tuning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [8] L. Li, D. Li, C. Lin, W. Li, W. Xue, S. Han, and Y. Guo, “AIRA: Activation-informed low-rank adaptation for large models,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [9] W. Lin, Y. Zhang, J. Dang, and L.-J. Zhang, “TLORA: Task-aware low rank adaptation of large language models,” *arXiv preprint arXiv:2604.18124*, 2026.
- [10] J. Liu, H. Kang, Q. Zhao, and G. Yu, “Training-free rank allocation for LoRA via representational sensitivity probing,” *arXiv preprint arXiv:2607.09757*, 2026.
- [11] Y.-A. Lee, C.-Y. Ko, P.-Y. Chen, and M.-Y. Yeh, “Learning rate matters: Vanilla LoRA may suffice for LLM fine-tuning,” *arXiv preprint arXiv:2602.04998*, 2026.
- [12] A. Aghajanyan, S. Gupta, and L. Zettlemoyer, “Intrinsic dimensionality explains the effectiveness of language model fine-tuning,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021, pp. 7319–7328.
- [13] J. He, C. Zhou, X. Ma, T. Berg-Kirkpatrick, and G. Neubig, “Towards a unified view of parameter-efficient transfer learning,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [14] N. Houlsby, A. Giurugu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, “Parameter-efficient transfer learning for NLP,” in *International Conference on Machine Learning (ICML)*, 2019.
- [15] B. Lester, R. Al-Rfou, and N. Constant, “The power of scale for parameter-efficient prompt tuning,” in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
- [16] X. L. Li and P. Liang, “Prefix-tuning: Optimizing continuous prompts for generation,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.
- [17] E. Ben Zaken, Y. Goldberg, and S. Ravfogel, “BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 1–9.

- [18] H. Liu, D. Tam, M. Muqeeth, J. Mohta, T. Huang, M. Bansal, and C. Raffel, "Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [19] S.-Y. Liu, C.-Y. Wang, H. Yin, P. Molchanov, Y.-C. F. Wang, K.-T. Cheng, and M.-H. Chen, "DoRA: Weight-decomposed low-rank adaptation," in *International Conference on Machine Learning (ICML)*, 2024.
- [20] D. J. Kopiczko, T. Blankevoort, and Y. M. Asano, "VeRA: Vector-based random matrix adaptation," in *International Conference on Learning Representations (ICLR)*, 2024.
- [21] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [22] D. Biderman, J. Portes, J. J. G. Ortiz, M. Paul, P. Greengard, C. Jennings, D. King, S. Havens, V. Chiley, J. Frankle, C. Blakeney, and J. P. Cunningham, "LoRA learns less and forgets less," *Transactions on Machine Learning Research*, 2024.
- [23] F. Meng, Z. Wang, and M. Zhang, "PiSSA: Principal singular values and singular vectors adaptation of large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [24] H. Wang, Y. Li, S. Wang, G. Chen, and Y. Chen, "MiLoRA: Harnessing minor singular components for parameter-efficient LLM finetuning," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025, pp. 4823–4836.
- [25] S. Wang, L. Yu, and J. Li, "LoRA-GA: Low-rank adaptation with gradient approximation," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [26] D. Kalajdziewski, "A rank stabilization scaling factor for fine-tuning with LoRA," *arXiv preprint arXiv:2312.03732*, 2023.
- [27] S. Hayou, N. Ghosh, and B. Yu, "LoRA+: Efficient low rank adaptation of large models," in *International Conference on Machine Learning (ICML)*, 2024.
- [28] S. Gururangan, A. Marasović, S. Swayamdipta, K. Lo, I. Beltagy, D. Downey, and N. A. Smith, "Don't stop pretraining: Adapt language models to domains and tasks," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 8342–8360.
- [29] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, "Domain-specific language model pretraining for biomedical natural language processing," *ACM Transactions on Computing for Healthcare*, vol. 3, no. 1, pp. 1–23, 2021.
- [30] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.
- [31] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A pretrained language model for scientific text," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- [32] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2016.
- [33] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *IEEE International Conference on Computer Vision (ICCV)*, 2013.
- [34] Z. J. Koles, M. S. Lazar, and S. Z. Zhou, "Spatial patterns underlying population differences in the background EEG," *Brain Topography*, vol. 2, no. 4, pp. 275–284, 1990.
- [35] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial EEG during imagined hand movement," *IEEE Transactions on Rehabilitation Engineering*, vol. 8, no. 4, pp. 441–446, 2000.
- [36] O. Kovaleva, S. Kulshreshtha, A. Rogers, and A. Rumshisky, "BERT busters: Outlier dimensions that disrupt transformers," in *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 2021.
- [37] G. Puccetti, A. Rogers, A. Drozd, and F. Dell'Orletta, "Outlier dimensions that disrupt transformers are driven by frequency," in *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022, pp. 1286–1304.
- [38] Z. Luo, A. Kulmizev, and X. Mao, "Positional artefacts propagate through masked language model embeddings," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021, pp. 5312–5327.
- [39] W. Timkey and M. van Schijndel, "All bark and no bite: Rogue dimensions in transformer language models obscure representational quality," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021, pp. 4527–4546.
- [40] Y. Bondarenko, M. Nagel, and T. Blankevoort, "Understanding and overcoming the challenges of efficient transformer quantization," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021, pp. 7947–7969.
- [41] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, "GPT3.int8(): 8-bit matrix multiplication for transformers at scale," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [42] M. Sun, X. Chen, J. Z. Kolter, and Z. Liu, "Massive activations in large language models," *arXiv preprint arXiv:2402.17762*, 2024.
- [43] C. Eckart and G. Young, "The approximation of one matrix by another of lower rank," *Psychometrika*, vol. 1, no. 3, pp. 211–218, 1936.
- [44] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.
- [45] B. Fox, "Discrete optimization via marginal analysis," *Management Science*, vol. 13, no. 3, pp. 210–216, 1966.
- [46] A. Federgruen and H. Groenevelt, "The greedy procedure for resource allocation problems: Necessary and sufficient conditions for optimality," *Operations Research*, vol. 34, no. 6, pp. 909–918, 1986.
- [47] M. Krallinger, O. Rabal, S. A. Akhondi *et al.*, "Overview of the BioCreative VI chemical-protein interaction track," *Proceedings of the BioCreative VI Workshop*, pp. 141–146, 2017.
- [48] F. Dernoncourt and J. Y. Lee, "PubMed 200k RCT: a dataset for sequential sentence classification in medical abstracts," in *Proceedings of the Eighth International Joint Conference on Natural Language Processing (IJCNLP)*, 2017, pp. 308–313.
- [49] S. Baker, I. Silins, Y. Guo, I. Ali, J. Högberg, U. Stenius, and A. Korhonen, "Automatic semantic classification of scientific literature according to the hallmarks of cancer," *Bioinformatics*, vol. 32, no. 3, pp. 432–440, 2016.
- [50] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv preprint arXiv:1907.11692*, 2019.
- [51] I. Turc, M.-W. Chang, K. Lee, and K. Toutanova, "Well-read students learn better: On the importance of pre-training compact models," *arXiv preprint arXiv:1908.08962*, 2019.
- [52] L. Ben Allal, A. Lozhkov, E. Bakouch *et al.*, "SmolLM2: When smol goes big – data-centric training of a small language model," *arXiv preprint arXiv:2502.02737*, 2025.
- [53] T. Wolf, L. Debut, V. Sanh *et al.*, "Transformers: State-of-the-art natural language processing," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2020, pp. 38–45.
- [54] S. Merity, C. Xiong, J. Bradbury, and R. Socher, "Pointer sentinel mixture models," in *International Conference on Learning Representations (ICLR)*, 2017.
- [55] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.
- [56] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [57] MTSamples, "Transcribed medical transcription sample reports and examples," <https://www.mtsamples.com>, specialty-labelled release galileo-ai/medical_transcription_40 on the Hugging Face Hub.

APPENDIX A

LEARNING-RATE SELECTION

For every configuration from which a conclusion is drawn, Table VIII lists the learning rate selected on the dev set (seed 1) and the range of the grid from which it was chosen.

TABLE VIII

LEARNING-RATE SELECTION (DEV SET, SEED 1) FOR ALL 119 TUNED CONFIGURATIONS. EACH GRID STARTS FROM THE LISTED FIRST VALUES AND IS EXTENDED ONE STEP PAST WHICHEVER EDGE HOLDS THE BEST SCORE UNTIL THE SELECTION IS INTERIOR; NO SELECTION LIES ON AN EDGE OF ITS FINAL GRID.

Configuration	Selected	Configuration	Selected
RoBERTa-base, ChemProt, DRIFT	3e-4 [1e-5, 1e-3]	RoBERTa-base, ChemProt, LoRA, $r=16$	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, LoRA	3e-4 [1e-4, 1e-3]	RoBERTa-base, ChemProt, EVA, $r=16$	3e-4 [3e-5, 1e-3]
RoBERTa-base, ChemProt, EVA	1e-4 [1e-5, 3e-4]	RoBERTa-base, ChemProt, EVA (whitened), $r=16$	3e-4 [1e-4, 1e-3]
RoBERTa-base, ChemProt, EVA (whitened)	1e-3 [1e-5, 3e-3]	BERT-Tiny, ChemProt, DRIFT	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, AdaLoRA	3e-3 [1e-4, 1e-2]	BERT-Tiny, ChemProt, LoRA	3e-3 [1e-4, 1e-2]
RoBERTa-base, ChemProt, Full fine-tuning	5e-5 [1e-5, 1e-4]	BERT-Tiny, ChemProt, EVA	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, BitFit	1e-3 [3e-4, 3e-3]	BERT-Tiny, ChemProt, EVA (whitened)	3e-3 [1e-4, 1e-2]
RoBERTa-base, ChemProt, Linear probe	5e-3 [1e-3, 2e-2]	BERT-Mini, ChemProt, DRIFT	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, PiSSA	3e-4 [1e-4, 1e-3]	BERT-Mini, ChemProt, LoRA	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, DoRA	1e-3 [1e-4, 3e-3]	BERT-Mini, ChemProt, EVA	1e-3 [1e-4, 3e-3]
RoBERTa-base, RCT-20k, DRIFT	1e-4 [1e-5, 3e-4]	BERT-Mini, ChemProt, EVA (whitened)	1e-3 [1e-4, 3e-3]
RoBERTa-base, RCT-20k, LoRA	3e-4 [1e-4, 1e-3]	BERT-Small, ChemProt, DRIFT	3e-4 [1e-4, 1e-3]
RoBERTa-base, RCT-20k, EVA	1e-4 [1e-5, 3e-4]	BERT-Small, ChemProt, LoRA	1e-3 [1e-4, 3e-3]
RoBERTa-base, RCT-20k, EVA (whitened)	1e-4 [1e-5, 3e-4]	BERT-Small, ChemProt, EVA	1e-3 [1e-4, 3e-3]
RoBERTa-base, RCT-20k, AdaLoRA	3e-3 [1e-4, 1e-2]	BERT-Small, ChemProt, EVA (whitened)	1e-3 [1e-4, 3e-3]
RoBERTa-base, RCT-20k, Full fine-tuning	3e-5 [1e-5, 5e-5]	BERT-Medium, ChemProt, DRIFT	3e-4 [1e-4, 1e-3]
RoBERTa-base, RCT-20k, BitFit	3e-3 [3e-4, 1e-2]	BERT-Medium, ChemProt, LoRA	1e-3 [1e-4, 3e-3]
RoBERTa-base, RCT-20k, Linear probe	2e-2 [1e-3, 5e-2]	BERT-Medium, ChemProt, EVA	3e-4 [1e-4, 1e-3]
RoBERTa-base, RCT-20k, PiSSA	1e-4 [3e-5, 3e-4]	BERT-Medium, ChemProt, EVA (whitened)	3e-4 [1e-4, 1e-3]
RoBERTa-base, RCT-20k, DoRA	1e-4 [3e-5, 3e-4]	BERT-Base, ChemProt, DRIFT	1e-3 [1e-4, 3e-3]
RoBERTa-base, HoC, DRIFT	3e-4 [1e-5, 1e-3]	BERT-Base, ChemProt, LoRA	3e-4 [1e-4, 1e-3]
RoBERTa-base, HoC, LoRA	3e-4 [1e-4, 1e-3]	BERT-Base, ChemProt, EVA	1e-4 [3e-5, 1e-3]
RoBERTa-base, HoC, EVA	3e-4 [1e-5, 1e-3]	BERT-Base, ChemProt, EVA (whitened)	1e-3 [1e-4, 3e-3]
RoBERTa-base, HoC, EVA (whitened)	1e-4 [1e-5, 3e-4]	SmolLM2-360M, HoC, LoRA	1e-3 [3e-4, 3e-3]
RoBERTa-base, HoC, AdaLoRA	3e-3 [1e-4, 1e-2]	SmolLM2-360M, HoC, EVA	1e-3 [1e-4, 3e-3]
RoBERTa-base, HoC, Full fine-tuning	5e-5 [1e-5, 1e-4]	SmolLM2-360M, HoC, EVA (whitened)	1e-3 [3e-4, 3e-3]
RoBERTa-base, HoC, BitFit	1e-3 [3e-4, 3e-3]	SmolLM2-360M, HoC, DRIFT	1e-3 [3e-4, 3e-3]
RoBERTa-base, HoC, Linear probe	2e-2 [1e-3, 5e-2]	RoBERTa-base, ChemProt, LoRA, $r=1$, rsLoRA scale	1e-3 [1e-4, 3e-3]
RoBERTa-base, HoC, PiSSA	3e-4 [1e-4, 1e-3]	RoBERTa-base, ChemProt, LoRA, $r=2$, rsLoRA scale	3e-4 [1e-4, 1e-3]
RoBERTa-base, HoC, DoRA	3e-4 [1e-4, 1e-3]	RoBERTa-base, ChemProt, LoRA, $r=4$, rsLoRA scale	3e-4 [1e-4, 1e-3]
RoBERTa-base, ChemProt, LoRA, attention only	1e-3 [1e-4, 3e-3]	RoBERTa-base, ChemProt, LoRA, rsLoRA scale	3e-4 [1e-4, 1e-3]
RoBERTa-base, ChemProt, LoRA, FFN only	1e-3 [1e-4, 3e-3]	RoBERTa-base, ChemProt, LoRA, $r=16$, rsLoRA scale	3e-4 [1e-4, 1e-3]
RoBERTa-base, ChemProt, LoRA, FFN only, $r=14$	1e-3 [1e-4, 3e-3]	RoBERTa-base, RCT-20k, DRIFT, $r=4$	3e-4 [1e-4, 1e-3]
RoBERTa-base, ChemProt, LoRA, $r=4$	3e-4 [1e-4, 1e-3]	RoBERTa-base, RCT-20k, EVA, $r=4$	1e-4 [3e-5, 3e-4]
RoBERTa-base, RCT-20k, LoRA, attention only	3e-4 [1e-4, 1e-3]	RoBERTa-base, RCT-20k, EVA (whitened), $r=4$	3e-4 [1e-4, 1e-3]
RoBERTa-base, RCT-20k, LoRA, FFN only	3e-4 [1e-4, 1e-3]	RoBERTa-base, RCT-20k, DRIFT, $r=16$	1e-4 [3e-5, 1e-3]
RoBERTa-base, RCT-20k, LoRA, FFN only, $r=14$	1e-3 [1e-4, 3e-3]	RoBERTa-base, RCT-20k, LoRA, $r=16$	3e-4 [1e-4, 1e-3]
RoBERTa-base, RCT-20k, LoRA, $r=4$	1e-3 [1e-4, 3e-3]	RoBERTa-base, RCT-20k, EVA, $r=16$	3e-4 [3e-5, 1e-3]
RoBERTa-base, HoC, LoRA, attention only	3e-4 [1e-4, 1e-3]	RoBERTa-base, RCT-20k, EVA (whitened), $r=16$	3e-4 [1e-4, 1e-3]
RoBERTa-base, HoC, LoRA, FFN only	1e-3 [1e-4, 3e-3]	RoBERTa-base, HoC, DRIFT, $r=4$	3e-4 [1e-4, 1e-3]
RoBERTa-base, HoC, LoRA, FFN only, $r=14$	1e-3 [1e-4, 3e-3]	RoBERTa-base, HoC, EVA, $r=4$	1e-4 [3e-5, 3e-4]
RoBERTa-base, HoC, LoRA, $r=4$	3e-4 [1e-4, 1e-3]	RoBERTa-base, HoC, EVA (whitened), $r=4$	1e-4 [3e-5, 1e-3]
RoBERTa-base, ChemProt, GEV	1e-3 [1e-4, 3e-3]	RoBERTa-base, HoC, DRIFT, $r=16$	1e-4 [3e-5, 1e-3]
RoBERTa-base, RCT-20k, GEV	1e-4 [3e-5, 1e-3]	RoBERTa-base, HoC, LoRA, $r=16$	3e-4 [1e-4, 1e-3]
RoBERTa-base, HoC, GEV	3e-4 [1e-4, 1e-3]	RoBERTa-base, HoC, EVA, $r=16$	1e-4 [3e-5, 3e-4]
RoBERTa-base, ChemProt, DRIFT, $\tau=0.5$	1e-3 [1e-4, 3e-3]	RoBERTa-base, HoC, EVA (whitened), $r=16$	3e-4 [1e-4, 1e-3]
RoBERTa-base, ChemProt, DRIFT, $\tau=0.9$	1e-3 [1e-4, 3e-3]	RoBERTa-base, ChemProt, LoRA, linear head	3e-4 [1e-4, 1e-3]
RoBERTa-base, ChemProt, DRIFT, $\tau=0.99$	1e-3 [1e-4, 3e-3]	RoBERTa-base, ChemProt, EVA, linear head	1e-4 [3e-5, 3e-4]
RoBERTa-base, ChemProt, DRIFT, $r=1$	3e-4 [1e-4, 1e-3]	RoBERTa-base, ChemProt, EVA (whitened), linear head	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, LoRA, $r=1$	1e-3 [1e-4, 3e-3]	RoBERTa-base, ChemProt, DRIFT, linear head	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, EVA, $r=1$	1e-4 [3e-5, 3e-4]	RoBERTa-base, ChemProt, BitFit, linear head	1e-3 [3e-4, 3e-3]
RoBERTa-base, ChemProt, EVA (whitened), $r=1$	1e-3 [1e-4, 3e-3]	RoBERTa-base, MTSamples, LoRA	1e-4 [3e-5, 1e-3]
RoBERTa-base, ChemProt, DRIFT, $r=2$	1e-3 [1e-4, 3e-3]	RoBERTa-base, MTSamples, EVA	1e-4 [3e-5, 3e-4]
RoBERTa-base, ChemProt, LoRA, $r=2$	1e-3 [1e-4, 3e-3]	RoBERTa-base, MTSamples, EVA (whitened)	1e-4 [3e-5, 1e-3]
RoBERTa-base, ChemProt, EVA, $r=2$	1e-4 [3e-5, 3e-4]	RoBERTa-base, MTSamples, DRIFT	1e-4 [3e-5, 1e-3]
RoBERTa-base, ChemProt, EVA (whitened), $r=2$	1e-3 [1e-4, 3e-3]	SmolLM2-360M, ChemProt, DRIFT	1e-3 [1e-5, 3e-3]
RoBERTa-base, ChemProt, DRIFT, $r=4$	1e-3 [1e-4, 3e-3]	SmolLM2-360M, ChemProt, EVA	3e-4 [1e-5, 3e-3]
RoBERTa-base, ChemProt, EVA, $r=4$	1e-4 [3e-5, 3e-4]	SmolLM2-360M, ChemProt, EVA (whitened)	1e-3 [1e-5, 3e-3]
RoBERTa-base, ChemProt, EVA (whitened), $r=4$	1e-3 [1e-4, 3e-3]	SmolLM2-360M, ChemProt, LoRA	1e-3 [1e-4, 3e-3]
RoBERTa-base, ChemProt, DRIFT, $r=16$	1e-3 [1e-4, 3e-3]		

APPENDIX B

PER-SEED RESULTS

Table IX lists the test score of every seed in Table I.

TABLE IX

PER-SEED TEST SCORES OF TABLE I (SEEDS 1, 2, 3 AND, FOR LoRA, EVA, WHITENED EVA AND DRIFT, SEEDS 4 AND 5), WITH THE 95% t -INTERVAL OF THE MEAN OVER SEEDS.

Method	ChemProt	95% CI	RCT-20k	95% CI	HoC	95% CI
Full fine-tuning	82.1 / 81.9 / 81.3	[80.7, 82.8]	84.0 / 84.1 / 83.0	[82.1, 85.3]	74.9 / 78.4 / 75.5	[71.5, 81.0]
Linear probe	49.4 / 48.7 / 49.8	[47.9, 50.7]	77.5 / 77.8 / 76.7	[75.9, 78.7]	48.5 / 50.9 / 47.0	[43.9, 53.7]
BitFit	81.4 / 81.4 / 81.2	[81.0, 81.7]	84.0 / 83.8 / 83.9	[83.6, 84.1]	77.5 / 78.7 / 78.5	[76.7, 79.8]
LoRA	82.6 / 82.1 / 82.7 / 82.5 / 82.7	[82.2, 82.8]	83.6 / 83.9 / 84.1 / 83.4 / 83.7	[83.4, 84.1]	76.5 / 78.3 / 78.9 / 76.6 / 79.5	[76.3, 79.7]
DoRA	81.9 / 82.2 / 83.0	[80.9, 83.9]	83.7 / 83.5 / 83.7	[83.3, 83.9]	80.5 / 14.3 / 79.1	[-36.0, 151.9]
PiSSA	82.1 / 82.0 / 83.1	[80.9, 83.9]	83.9 / 83.8 / 83.8	[83.7, 84.0]	61.5 / 14.3 / 77.8	[-30.7, 133.1]
AdaLoRA	82.6 / 81.8 / 83.6	[80.5, 84.9]	84.3 / 82.0 / 80.7	[77.8, 86.9]	79.0 / 77.7 / 77.7	[76.3, 79.9]
EVA (budget-matched)	82.7 / 80.0 / 82.7 / 80.5 / 81.5	[80.0, 83.0]	83.9 / 83.5 / 83.7 / 83.9 / 83.7	[83.5, 84.0]	77.5 / 26.2 / 76.5 / 79.0 / 37.8	[27.9, 90.9]
EVA (whitened)	82.9 / 82.8 / 82.2 / 83.8 / 78.5	[79.5, 84.6]	83.9 / 83.8 / 84.0 / 84.0 / 84.1	[83.8, 84.1]	78.0 / 77.8 / 76.4 / 78.8 / 75.5	[75.6, 78.9]
DRIFT	82.1 / 82.1 / 82.8 / 82.5 / 82.5	[82.0, 82.8]	84.0 / 84.0 / 83.9 / 83.7 / 84.0	[83.8, 84.1]	61.4 / 79.0 / 79.3 / 80.0 / 78.7	[65.8, 85.6]
GEV	81.8 / 82.9 / 82.6	[81.0, 83.8]	83.9 / 83.9 / 84.0	[83.8, 84.0]	75.2 / 79.4 / 78.3	[72.2, 83.0]

APPENDIX C

INSTANCE-LEVEL INTERVALS

The runs of Table I did not store per-example predictions, so we repeated every one of them with the same configurations, rates and seeds, this time storing the predictions, and we bootstrap over seeds and test examples jointly (Table X). Paired differences to LoRA resample the same seeds and examples for both methods. The replicate means also measure run-to-run variation under fp16. On ChemProt and RCT-20k every replicate reproduces its Table I run exactly, while on HoC the two differ by 0.5 to 21 points. The large values come from seeds that failed in one run and not in the other (DoRA +21.1, PiSSA +11.8, EVA +6.0, DRIFT +3.3, against +1.0 for LoRA).

TABLE X

INSTANCE-LEVEL UNCERTAINTY FOR TABLE I: A REPLICATION OF EVERY PARAMETER-EFFICIENT RUN THAT STORES PER-EXAMPLE TEST PREDICTIONS, WITH 95% HIERARCHICAL BOOTSTRAP INTERVALS (SEEDS, THEN TEST EXAMPLES, 2,000 REPLICATES). Δ : PAIRED DIFFERENCE TO LoRA (SAME SEEDS AND EXAMPLES). REP.: REPLICATE MEAN MINUS THE TABLE I MEAN (FP16 RUN-TO-RUN VARIATION).

Method	ChemProt			RCT-20k			HoC		
	Mean [95% CI]	Δ vs LoRA [95% CI]	Rep.	Mean [95% CI]	Δ vs LoRA [95% CI]	Rep.	Mean [95% CI]	Δ vs LoRA [95% CI]	Rep.
BitFit	81.3 [80.1, 82.5]	-1.1 [-2.0, -0.2]	+0.0	83.9 [82.9, 84.8]	+0.0 [-0.6, +0.6]	+0.0	79.5 [76.2, 82.8]	+0.6 [-1.9, +3.3]	+1.3
LoRA	82.5 [81.3, 83.7]	—	+0.0	83.7 [82.8, 84.6]	—	+0.0	79.0 [75.7, 82.3]	—	+1.0
DoRA	82.4 [81.1, 83.6]	-0.1 [-1.0, +0.8]	+0.0	83.6 [82.7, 84.5]	-0.2 [-0.7, +0.3]	+0.0	79.0 [75.8, 82.1]	+0.1 [-2.5, +2.6]	+21.1
PiSSA	82.4 [81.1, 83.7]	-0.1 [-0.9, +0.7]	+0.0	83.8 [82.9, 84.7]	-0.1 [-0.6, +0.5]	+0.0	63.0 [34.6, 80.6]	-15.9 [-45.9, +1.9]	+11.8
AdaLoRA	82.7 [81.3, 84.1]	+0.2 [-0.7, +1.1]	+0.0	82.3 [80.5, 84.2]	-1.5 [-3.4, +0.5]	-0.0	78.8 [75.4, 82.0]	-0.1 [-2.8, +2.6]	+0.7
EVA (budget-matched)	81.5 [80.0, 83.0]	-1.0 [-2.1, +0.0]	+0.0	83.7 [82.8, 84.7]	+0.0 [-0.5, +0.6]	+0.0	65.4 [39.5, 80.6]	-13.6 [-41.3, +1.3]	+6.0
EVA (whitened)	82.0 [80.1, 83.9]	-0.5 [-2.5, +1.1]	+0.0	84.0 [83.0, 84.8]	+0.2 [-0.2, +0.7]	-0.0	77.8 [74.3, 80.9]	-1.2 [-3.4, +0.9]	+0.5
DRIFT	82.4 [81.3, 83.6]	-0.1 [-0.7, +0.5]	+0.0	83.9 [83.0, 84.8]	+0.2 [-0.3, +0.7]	+0.0	79.0 [75.8, 82.2]	+0.0 [-2.0, +2.1]	+3.3
GEV	82.4 [81.1, 83.6]	-0.1 [-1.1, +1.0]	+0.0	83.9 [83.0, 84.8]	+0.1 [-0.5, +0.6]	+0.0	79.7 [76.4, 82.9]	+0.8 [-1.1, +3.0]	+2.1

APPENDIX D

A LINEAR CLASSIFICATION HEAD

Every method trains RoBERTa’s classification head, whose 0.60M parameters are shared by all methods and could compress their differences. Replacing the head with a single linear layer on the first token, and tuning every configuration again, lowers every method except DRIFT by 0.6–1.5 points and changes no conclusion (Table XI). LoRA reaches 81.9 ± 0.1 , DRIFT 82.7 ± 0.7 ($+0.8$, $p = 0.14$), whitened EVA 81.1 ± 0.8 , BitFit 80.6 ± 0.5 and EVA 80.4 ± 0.3 (-1.5 , $p = 0.03$ uncorrected).

TABLE XI

CHEMPROT WITH THE BACKBONE’S CLASSIFICATION HEAD (TABLE I, SEEDS 1–3) AND WITH A SINGLE LINEAR HEAD ON THE FIRST TOKEN (768×13), EVERY CONFIGURATION TUNED ON ITS OWN (TEST MICRO-F1, THREE SEEDS).

Method	RoBERTa head (0.60M)	Linear head (0.01M)
LoRA	82.5 ± 0.3	81.9 ± 0.1
EVA (budget-matched)	81.8 ± 1.5	80.4 ± 0.3
EVA (whitened)	82.6 ± 0.4	81.1 ± 0.8
DRIFT	82.3 ± 0.4	82.7 ± 0.7
BitFit	81.3 ± 0.1	80.6 ± 0.5

APPENDIX E

τ -FREE DIVERGENCES

Table XII gives three divergences between the input covariances of each corpus and those of WikiText. Unlike the drift ratio, these do not depend on τ . They are the CORAL distance, the Bures–Wasserstein distance and the symmetrised log-det (Jeffreys) divergence between trace-normalised covariances, each averaged over the 48 distinct input sites. The floor is the divergence between two disjoint halves of WikiText at the same sequence length.

TABLE XII
TAU-FREE DIVERGENCE OF EACH CORPUS’S INPUT COVARIANCES FROM THE WIKITEXT REFERENCE (TRACE-NORMALISED; MEAN OVER THE 48 DISTINCT INPUT SITES).
FLOOR: A DISJOINT HALF OF WIKITEXT AT THE SAME SEQUENCE LENGTH. NEWS AND RANDOM TOKENS AT CHEMPROT’S LENGTH.

Corpus vs. WikiText	CORAL	Bures	Jeffreys
ChemProt	1.172	0.3456	0.653
floor (ChemProt length)	0.076	0.0068	0.016
News (CNN/DailyMail)	0.920	0.1787	0.281
Random tokens	1.058	0.4378	0.893
RCT-20k	0.878	0.2306	0.411
floor (RCT-20k length)	0.081	0.0079	0.019
HoC	0.763	0.2093	0.393
floor (HoC length)	0.075	0.0056	0.013
MTSamples	1.105	0.2770	0.495
floor (MTSamples length)	0.075	0.0056	0.013

APPENDIX F

BUDGETS ON RCT-20K AND HoC

Table XIII extends the budget comparison of Fig. 3 to RCT-20k and HoC at ranks 4 and 16, with every method and budget tuned on its own. Rank 8 repeats the configuration of Table I over seeds 1–3.

TABLE XIII
ADAPTER BUDGET ON RCT-20K AND HoC (MEAN $\pm$ S.D., THREE SEEDS, EVERY METHOD AND BUDGET TUNED ON ITS OWN). RANK 4, 8 AND 16 SPEND 0.53%, 1.06% AND 2.1% OF THE BACKBONE.

Task	r	LoRA	EVA	EVA (whitened)	DRIFT
RCT-20k	4	83.5 $\pm$ 0.5	83.8 $\pm$ 0.2	83.9$\pm$ 0.1	83.9$\pm$ 0.4
	8	83.9 $\pm$ 0.2	83.7 $\pm$ 0.2	83.9 $\pm$ 0.1	84.0$\pm$ 0.1
	16	84.0 $\pm$ 0.1	83.5 $\pm$ 0.3	84.1$\pm$ 0.1	83.9 $\pm$ 0.2
HoC	4	78.7$\pm$ 1.0	77.4 $\pm$ 1.6	77.4 $\pm$ 0.6	61.3 $\pm$ 29.1
	8	77.9$\pm$ 1.3	60.1 $\pm$ 29.3	77.4 $\pm$ 0.9	73.2 $\pm$ 10.2
	16	80.3$\pm$ 1.6	77.8 $\pm$ 2.2	79.0 $\pm$ 0.5	74.9 $\pm$ 0.6